Olympiad Combinatorics

A Chapter-by-Chapter Study Guide

Companion notes to

Pranav A. Sriram, *Olympiad Combinatorics* (August 2014)

What this document is

These are *study notes*, not a substitute for the book. The book is a nine-chapter working text: each chapter develops a family of techniques through motivated, fully worked olympiad problems, then closes with a large graded exercise set. These notes do three things the book deliberately does not do all in one place:

- they **compress each chapter to its skeleton** — the one central idea, the tools it installs, and how those tools are recognised in a fresh problem;
- they **index the reusable lemmas** scattered across nine chapters into a single reference table (Section 11);
- they **map problems to methods** so that, facing a new problem, you can ask "which chapter does this belong to?" instead of "which trick was that?"

The worked examples of the book are named and their *method* is summarised — read the source for the full write-ups; the value added here is the *pattern*.

Prerequisites and how to read this

Prerequisites. Comfort with induction, basic set and graph language (vertices, edges, degree, tree, clique, colouring), binomial coefficients, and — for **Ch. 9** only — high-school probability (random variables, expectation, variance, independence). No real analysis is needed.

How to read. Skim Section first to see the shape of the book. Then read the chapter notes in order: **Ch. 1–4** build the constructive and existential core; **Ch. 5–6** are largely independent (games; double counting) but lean on **Ch. 1–4**; **Ch. 7–9** are the "deep dive" and reuse everything. Each chapter note has the same five parts — *Snapshot*, *Essential idea*, *Tools and results*, *Landmark problems*, *Takeaways and pitfalls* — so you can use them as a revision checklist before a contest.

Source text: Pranav A. Sriram, *Olympiad Combinatorics*, August 2014 (9 chapters).
All problem attributions in these notes (IMO, ISL, USAMO, Russia, China TST, ELMO, . . .)
are the book's own attributions.

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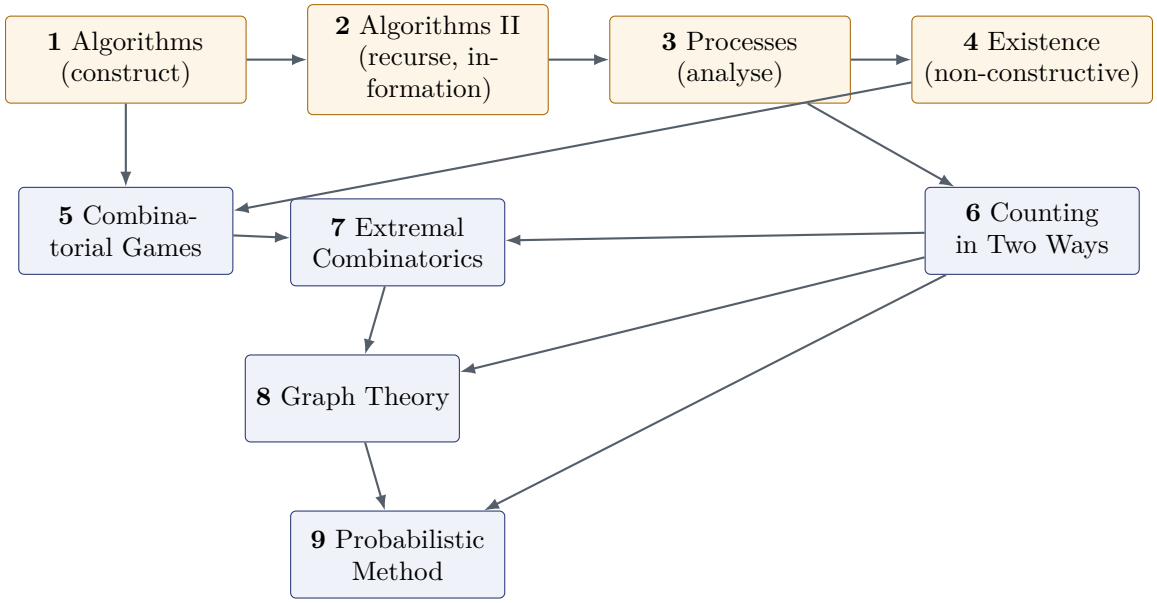
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How the book is built

Sriram's book is organised by *method*, not by problem topic. That is the single most important structural fact about it. A "graph problem" can appear in Chapter 6 (count it twice), Chapter 7 (bound the extremal case), Chapter 8 (apply a structural lemma) or Chapter 9 (show a random object works). The nine chapters are three concentric arcs, each arc reusing the previous one.



Arc 1 — Construct and analyse (Ch. 1–Ch. 4). The first four chapters are one long argument about *how to prove a combinatorial statement*. **Ch. 1** builds objects with greedy procedures and controls them with invariants and monovariants. **Ch. 2** adds recursion: divide the task, or reduce n to $n-1$, and count *information* rather than objects. **Ch. 3** turns the camera around and analyses processes someone else designed — does it stop, where does it stop, how long does it take. **Ch. 4** then drops construction entirely and teaches non-constructive existence: extremal principle, infinite descent, optimal assumption, hostile neighbours, discrete continuity.

Arc 2 — Compete and count (Ch. 5–Ch. 6). **Ch. 5** is the only chapter with an adversary: symmetry/copying, pairing, and positional analysis (P- and N-positions). **Ch. 6** is the workhorse: compute one quantity two ways, get an identity or an inequality. Almost every hard inequality in **Ch. 7** and **Ch. 8** is ultimately a double count.

Arc 3 — Deep combinatorics (Ch. 7–Ch. 9). **Ch. 7** is extremal combinatorics proper — injections, alternating chains, classification, Kleitman's lemma, the Sunflower lemma. **Ch. 8** is graph theory as a modelling language: dominating sets, spanning trees, the extremal principle in graphs, Hall's marriage theorem, and graphs built for problems that look graph-free. **Ch. 9** replaces explicit construction by randomness: linearity of expectation, mutually exclusive events, bounding violated conditions, concentration inequalities, and the Lovász Local Lemma.

#	Core question	Central tool	Feeds into
1	How do I *build* the object?	Greedy; invariants & monovariants	3, 5
2	How do I build it *cheaply*?	Recursion; information/decision counting	4, 8
3	What happens if I just *run* it?	Invariant, monovariant, good/bad objects, step bounds	4, 5, 7
4	Does it exist at all?	Extremal principle, infinite descent, optimal assumption, hostile neighbours, discrete continuity	7, 8, 9
5	Who wins?	Symmetry/copying, pairing, P/N-positions	7
6	Can I count it *differently*?	Double counting, incidence matrices, degree identities	7, 8, 9
7	What is the maximum/minimum?	Injection, alternating chains, classification, set-induction, Sunflower	8, 9
8	What is the *structure*?	Dominating sets, spanning trees, extremal principle, Hall, modelling	9
9	Does something exist when nothing is constructible?	Expectation, union bound, concentration, LLL	—

Part I Construct and Analyse

1. Algorithms

Snapshot

An *algorithm* is a procedure designed to accomplish a task. **Ch. 1** uses algorithms not to compute but to *prove*: if you can write down an explicit procedure that produces the required object, you have proved the object exists. This is a *constructive proof*, and it is the most common way olympiad combinatorics proves an "exists" statement.

The chapter has three movements: **greedy algorithms** (make the locally best choice at each step); **invariants and monovariants** (find a quantity that does not change, or changes only one way, and use it both to *decide what to do next* and to *prove the procedure stops*); and a set of **miscellaneous constructions** where the trick is a good observation plus patience.

The essential idea

A constructive proof has two halves that are always separate, and both must be written: **(i) the procedure** — what do I do at each step? and **(ii) the certificate** — why does it terminate, and why is the output correct? Almost every lost mark on an algorithm problem is half (ii) missing or hand-waved. The book's own habit is to state a *heuristic* first ("the smallest uncovered number is the most unsafe, so cover it first"), and only then turn the heuristic into an algorithm — the heuristic is what makes the algorithm findable in the first place.

Tools and results

(a) Greedy algorithms and the exchange argument

A greedy algorithm maximises short-term gain. It is not always optimal, but when it is, it is usually proved optimal by an *exchange argument*: suppose an optimal solution differs from the greedy one; swap the offending piece; show the objective does not get worse. In the book's problems the exchange step is often disguised as "look at an *extreme* element" — the largest, smallest, best or worst object in a finite set.

Three canonical shapes recur:

- **Greedy assignment.** Order the objects, then process them in that order making the locally best legal choice. *Example:* colour the vertices of a graph of maximum degree Δ in any order, always using the smallest colour not used on an already-coloured neighbour. Since a vertex has at most Δ neighbours, one of $\{1, \dots, \Delta + 1\}$ is always free: $\chi(G) \le \Delta + 1$.
- **Greedy deletion.** Keep deleting the "bad" objects while a monovariant improves. *Example:* delete any vertex of degree $< E/V$. Each deletion removes fewer than E/V edges while removing exactly one vertex, so the ratio #edges/#vertices *strictly increases*; it cannot reach the value 0 of a single-vertex graph, so the process stops with all degrees $\ge E/V$. That is: **a graph of average degree $\bar{d}$ has an induced subgraph of minimum degree at least $\bar{d}/2$.**
- **Greedy with a priority order.** Choose the order so that the hard constraint is met. *Example:* cover the non-negative integers by *historic sets* $\{x, x+a, x+a+b\}$ or $\{x, x+b, x+a+b\}$ (ISL 2001 C4, $a = 1776$, $b = 2001$): always take x to be the *smallest uncovered* integer, and prefer the "small" set. The proof that it never gets stuck is a one-paragraph contradiction using "smallest" twice.

(b) Invariants and monovariants

An *invariant* is a quantity unchanged by every legal move; a *monovariant* is one that changes monotonically. Their division of labour:

- **The invariant proves impossibility.** If I is invariant, the start and the end must have equal I — a cheap necessary condition. *Example:* adding k to two adjacent cells of a chessboard always adds to one black and one white cell, so $S_{\text{black}} - S_{\text{white}}$ is invariant; reducing everything to 0 requires $S_{\text{black}} = S_{\text{white}}$ initially (ISL 1989). *Example:* a "parity" invariant proves the answer is negative.
- **The monovariant proves termination and directs the moves.** If every move strictly decreases a positive-integer quantity, the process must stop. The book's repeated slogan: *the monovariant answers the question "well, what do I do now?"*
- **Weighting is how monovariants are manufactured.** Assign to each object a weight so that a move has a controlled effect. Two devices pay off again and again:
  - *Geometric weights.* Pebble at (i, j) gets weight $2^{-(i+j)}$; some fixed fraction is consumed per move; the total is compared with the weight of the whole region (Indian TST 2004).
  - *Positional weights.* A coin with girl i gets weight i; a move changes the total by 0 or $\pm n$, so the total is invariant modulo n (ISL 1994 C5). Or: a cookie at distance i from the target gets weight 2^{-i}, and cookies passing *towards* the target preserve total weight while cookies passing away decrease it (New Zealand IMO Training 2011).
- **Sum-of-absolute-values monovariant.** For redistributing coins around a circle to make everyone equal, the quantity $X = |d_1| + |d_1 + d_2| + \cdots + |d_1 + \cdots + d_{n-1}|$, where $d_i = c_i - k$ is the surplus of person i, changes by exactly ± 1 under a move between positions j and $j + 1$. So each move can be chosen to reduce X by one, and X counts the *minimum* number of moves (APMO 1997 P5).

(c) Good/bad decomposition of a position

A recurring trick for very hard constructive problems: split positions into "good" (the monovariant can be reduced) and "bad". Write

$$\textit{sub-algorithm } A : \text{bad} \longrightarrow \text{good} \quad \text{(without worsening the monovariant)},$$
$$\textit{sub-algorithm } B : \text{good} \longrightarrow \text{strictly better}.$$

Then alternate A and B. *Flagship:* USAMO 2003-6 (six non-negative integers, sum 2003, at the vertices of a hexagon; a move replaces a vertex by |difference of its neighbours|). Sub-algorithm A: from odd sum to *exactly one odd entry*, using the parity pattern of $A + C + E$; sub-algorithm B: from one odd entry, decrease the maximum (the cases split on whether the maximum is even, and if odd, on whether $C = E = 0$). Because the sum stays odd and the maximum strictly decreases, the process reaches all zeros.

Landmark problems from Ch. 1

Problem	Method in one line
Colouring, Δ bound	Greedy: smallest free colour; only Δ neighbours can block.
$2\times n$ array, column sums 1 (Russia 2005)	Sort the top row, greedily take the smallest entries while the budget allows; the last unselected top entry is at least an average, which bounds the largest bottom entry. *Ordering the variables creates structure.*
Induced subgraph, $\deg \geq E/V$	Greedy deletion; monovariant = ratio edges/vertices, strictly increasing.
Historic sets (ISL 2001 C4)	Heuristics first (smallest uncovered x is the most unsafe; prefer small sets), then a contradiction from "$x_{n+1}+a$ and $x_{n+1}+b$ both already covered".
Chessboard, add k to adjacent pair (ISL 1989)	Invariant $S_{\text{black}} - S_{\text{white}}$; sufficiency by a two-cell reduction algorithm ($a \leq b$, then $a \geq b$ cases) applied row by row, then column by column.
$2n$ people, m cookies (NZ 2011)	Weight cookie at distance i by 2^{-i}; total weight is a monovariant. Threshold $m \geq 2^n$; then a refined monovariant W^+ on the better semicircle.
Peter's three bank accounts (ISL 1994 C3)	Strictly decrease $\min(A,B,C)$; Euclidean algorithm $B = qA + r$; write q in binary and let the doubling implement Aq.
n people, nk coins in a circle (APMO 1997 P5)	The $X = \sum \lvert s_j \rvert$ monovariant; each move reduces X by 1, so X_0 moves are necessary and sufficient.
Hexagon, USAMO 2003-6	Good/bad positions: create exactly one odd entry, then strictly decrease the maximum; sum stays odd.
Cards at $A_1, \ldots, A_n, O$ (China 2010 P5)	Observations (a)–(d): equalise, reduce to entries in $\{0,1,2\}$, shift to $\{1,2,3\}$, then arrange that every two 3s are separated by a 1.
Six boxes, IMO 2010 P5	Two sub-algorithms generating *towers of 2s*: $(a,0,0) \to (0,2^a,0)$ and then $(a,0,0,0) \to (0,P_a,0,0)$ where P_a is a power tower. Feed the tower into B_4.
Two rooms, IMO 2007 P3	Move one person at a time from the room containing a maximum clique; $c(A)$ drops by 1 while $c(B)$ rises by at most 1, so $c(B) - c(A) \in \{0,1\}$; finish with the "no competitor in $B \cap M$ outside the biggest B-clique" case.

Key takeaways

- **Extremes first.** When stuck, look at the largest, smallest, best or worst object. Almost every greedy proof in this chapter begins there.
- **An ordering is a tool.** Sorting a row, or processing vertices in a clever order, converts an unstructured statement into a structured one.
- **Monovariants do two jobs.** They tell you *what to do next* and they *prove termination* (a strictly decreasing positive integer cannot descend for ever).
- **Invariants are cheap necessary conditions.** They rarely finish a problem alone, but they kill false answers instantly.
- **Manufacture the weight, don't guess it.** Ask: "what assignment makes this move's effect clean?" Doubling moves want $2^{\pm i}$ weights; cyclic moves want mod$n$ weights.
- **Two sub-algorithms beat one clever algorithm.** If a single monovariant gets stuck on some positions, write a second routine that only fixes those positions, and alternate.
- **Towers, recursion and huge numbers are a hint.** When the target is astronomically large

(IMO 2010 P5: $2010^{2010^{2010}}$), the intended construction *generates* size recursively rather than moves it.

Where it usually goes wrong

- **Greedy without the exchange argument.** "Always take the biggest" is not a proof. Either give an exchange argument, or give a monovariant.
- **A monovariant that only "usually" decreases.** Check every legal move, including the ones you did not think of. A move that leaves the quantity the same is fine; one that increases it kills the argument.
- **Forgetting that the process must be able to make progress at all.** "X decreases at every step" plus "$X \geq 0$" is enough; "X decreases" alone is not, if you never show a decreasing step exists.
- **Proving the necessary condition and stopping.** Half of these problems ask for "necessary *and* sufficient".
- **Assuming the target value is reachable because it is huge.** IMO 2010 P5 looks impossible; the answer is yes, but only after you build the tower machinery.

2. Algorithms — Part II

Snapshot

Ch. 2 adds the two things **Ch. 1** was missing: **recursion** and **efficiency**. A recursive algorithm solves a problem on n objects by solving one or two *smaller* problems of the same kind — either $n-1$ (induction) or $n/2$ (divide and conquer) — and then combining. The second half of the chapter is about *information*: how many questions, comparisons, weighings or operations are needed to determine something, and how to prove that you cannot do better.

The essential idea

The two halves are joined by one idea: **the size of the answer set is the currency**. Reducing n to $n/2$ costs one "halving"; reducing n to $n-1$ costs one "step". So:

$$\text{linear target} \implies \text{relate } f(n) \text{ to } f(n-1),$$
$$\text{logarithmic target} \implies \text{relate } f(n) \text{ to } f(n/2) \text{ or use binary representations.}$$

Read the bound you are asked to prove — it tells you which recursion to set up. The book states this explicitly in the remark after the card-reversing problem: $2n-6$ hints at $f(n) \leq f(n-1)+2$; $\lceil \log_2 n \rceil$ hints at $f(n) = f(n/2)+1$.

Tools and results

(a) Induction done properly

- **Ignore one object, but pay for it.** China Girls MO 2011-7: n boxes in a row, move a ball from B_1 to B_2 or from B_n to B_{n-1}, or replace two balls in B_k by one in each neighbour. Plan: first *force a ball into* B_n (using the weight $\sum a_k 2^k$ monovariant, whose growth is bounded), then apply the $(n-1)$-box induction. The induction step needs a repair: the algorithm on $n-1$ boxes uses B_{n-1} as an edge box, but now B_{n-1} is interior and needs two balls to act — so *carry a coloured "blue" ball* and shuffle it out of the way whenever the induction wants to move from B_{n-1}.
- **Local repair instead of global restart.** ISL 2005 C1 (rooms of lamps, each lamp shares a switch with one other): call a room *bad* if all its lamps agree. Switch a lamp in a bad room; the badness moves to another room. Follow the trail of rooms. Either you make a room good without creating a new bad room, or you revisit a room — and a revisit is fine because (i) the two visits switched *different* lamps, and (ii) each room has ≥ 3 lamps, so the room is good. Hence each "tour" reduces

the number of bad rooms by one.

- **Induct by extending a subgraph.** A graph with minimum degree $\geq n-1$ contains a copy of *every* n-vertex tree T: delete a leaf v of T, find $T \setminus v$ by induction, and attach v to a neighbour of the image of T's neighbour that is not already used — one exists because the degree is $\geq n-1$ and at most $n-2$ vertices are used.
- **Induct on the right parameter.** USAMO 2002: colour all subsets of a 2002-element set black/white so that white is closed under union, black is closed under union, and exactly N white sets, for $0 \leq N \leq 2^{2002}$. The tempting induction is on N; the working induction is on $|S|$, colouring all *new* subsets black when $N \leq 2^{n-1}$, and swapping the roles of black and white when $N > 2^{n-1}$.

(b) Divide and conquer and the $n \log n$ profile

Merge sort: split into two halves, sort each, merge. The merge of two sorted lists costs at most n comparisons, so

$$f(n) \leq 2f(n/2) + n \implies f(n) = O(n \log_2 n),$$

against the naive $n(n-1)/2$. *Moral:* the recursion $f(n) = af(n/b) + cn$ with $a = b$ gives an extra logarithmic factor; with $a < b$ it collapses to linear; with $a > b$ it becomes polynomial in n.

(c) Information and decision counting

- **Binary labelling turns a search into a vector of answers.** n lamps and n switches, unknown matching; one operation flips a chosen set of switches simultaneously. Write each switch's label in binary and, in operation k, flip all switches with a 1 in the k-th binary position. Each lamp's pattern of "came on in operations $2, 3, 5$" *is* its switch's binary label. Total: $\lceil \log_2 n \rceil$ operations. Note the recursive reading of the same algorithm: $f(n) = f(n/2) + 1$, because the two halves can be processed *simultaneously*.
- **Ask for a modular invariant to read off bits.** Russia 2000: guess $X \leq 100$ by asking for $\gcd(X+M, N)$ with $M, N < 100$. Ask $\gcd(X+2, 2)$ to get the parity; if X is even ask $\gcd(X+4, 4)$, else $\gcd(X+1, 4)$; in general, knowing $X \equiv i \pmod{2^k}$, ask $\gcd(X + (2^k - i), 2^{k+1})$ to decide the next bit. Seven questions because $2^6 < 100 < 2^7$.
- **Adversarial counting gets constants.** Russia 2004: n boxes, an even positive number of white balls; a question names two boxes and asks whether at least one is white. Ask $(1, j)$ for $j = 2, \ldots, n$; then $(2, j)$ for $j = 3, \ldots, n$: $2n-3$ questions find a black box or certify two white ones. The parity hypothesis is used exactly once, to rule out "box 1 black, all others white".
- **Binary search with one lie.** Guessing $X \leq 2^k$ costs k questions without lying, and $2k+1$ by the naive "ask everything twice". But asking the k binary digits, then "did you ever lie?" and "was the previous answer a lie?", reduces the problem to *locating which of k answers was the lie* — a binary search for a number $\leq k$. Total $\lceil \log_2 n \rceil + \lceil \log_2 \lceil \log_2 n \rceil \rceil + 2$.

(d) One hard case: induction plus the extremal principle

IMO 2009 P6 (grasshopper jumps $a_1 < \cdots < a_{n+1}$ from 0, must avoid a set M of n forbidden points). Induct on n; let $m = \min M$ and split on $m < a_{n+1}$ or $m \geq a_{n+1}$. In the first case, either jump a_{n+1} immediately (and reduce to $M \setminus \{m\}$), or note that among the n pairs $(a_i, a_i + a_{n+1})$ at least one, say $(a_k, a_k + a_{n+1})$, has *both* members outside M — then two jumps clear two forbidden points at once. In the second case, run the problem *backwards from s*; if a jump lands on m, swap that jump for a_n and jump past everything.

Landmark problems from Ch. 2

Problem	Method in one line
Boxes and balls (CGMO 2011-7)	Force a ball into B_n via a weight monovariant, then induct; repair the edge-box problem with a movable "blue ball".
Lamps and rooms (ISL 2005 C1)	Chase the bad room around a cycle; the first repeat is harmless because two distinct lamps were switched and the room has ≥ 3 lamps.
Tree inside a dense graph	Induction on T by deleting a leaf; degree $\geq n-1$ guarantees a spare neighbour.
Black/white union-closed colouring (USAMO 2002)	Induct on $\lvert S\rvert$, not on N.
Merge sort	$f(n) \leq 2f(n/2) + n$.
Lamps and switches ($\lceil \log_2 n \rceil$)	Binary labels, or $f(n) = f(n/2) + 1$ with both halves processed at once.
Block reversal (gen. ISL 1998 C3)	$f(n) \leq f(n-1) + 2$; verify $f(5) \leq 4$ by hand; answer $2n - 6$.
gcd guessing (Russia 2000)	Read the binary digits of X one at a time with moduli $2, 4, 8, \ldots$
White-ball pairs (gen. Russia 2004)	$2n - 3$ questions; parity rules out the last degenerate case.
Binary search, and with one lie	Halving; then locate the lie by searching for its index.
Grasshopper (IMO 2009 P6)	Induction on n plus an extremal choice ($m = \min M$); two cases, the second run in reverse.

Key takeaways

- **Let the target bound choose the recursion.** Linear → subtract one; logarithmic → halve; if the problem mentions 2^n objects, look for a way to split the set in half.
- **Ignoring an object is a technique — but budget for it.** Keep a "parking spot" (a blue ball, a spare vertex, a leaf) to absorb the interference.
- **Divide and conquer is about *simultaneous* sub-problems.** $f(n) = 2f(n/2) + 1$ and $f(n) = f(n/2) + 1$ differ by a factor of n; look for independence between the halves before you write the recursion.
- **Information = number of possible answers.** To prove a lower bound on questions, count outcomes per question and multiply. To prove an upper bound, design an encoding (binary labels are the standard one).
- **Exhaust the easy cases first.** In IMO 2009 P6 and many others, the easy case is not just "out of the way" — solving it tells you the structure of the hard case.
- **Induct on the parameter that the hypothesis constrains.** USAMO 2002 inducts on the size of the ground set even though N is the parameter in the statement.

Where it usually goes wrong

- **Wrong induction parameter, or a base case quietly assumed.** USAMO 2002 refutes the natural induction; the block-reversal problem needs $n \geq 5$ before $f(n) \leq f(n-1) + 2$ gives the advertised bound.
- **Recursion with an unproved "combining" step.** The split is easy; the merge (or the repair) is the problem. Write it out.
- **Confusing $\lceil \log_2 n \rceil$ with $\lfloor \log_2 n \rfloor$** when n is a power of two or one more than a power of two; these are exactly the cases graders check.
- **Lower bounds by "a question has two answers so we need $\log_2(\cdot)$" without checking**

that all answers are achievable. A decision-tree lower bound needs the set of possible answers to be distinguishable by the available questions.

3. Processes

Snapshot

Ch. 1 and **Ch. 3** both feature an initial configuration and a set of allowed moves, but the question is inverted. In **Ch. 1** *you* design the procedure to reach a target. In **Ch. 3** someone hands you a process and asks:

- Does it terminate?
- If it terminates, what does the final configuration look like? Is it unique?
- How many steps can it take before a particular configuration occurs?

The chapter's tool kit is invariants, the extremal principle, induction, and a beautiful idea the book calls **problem alteration** — changing the rules of the process in a way that does not change the answer, because the new process is easier to analyse. The book also insists on the unglamorous part: *experiment with small values, observe, conjecture, then prove.*

The essential idea

Three questions, three reflexes.

Is it impossible?	Find an invariant: a weight, a parity, a value mod$m$.
Does it stop?	Find a monovariant with values in a well-ordered set, or find a quantity that can only be changed finitely often.
What is the end state?	Take the *earliest* configuration with the bad property and look at the move immediately before it — a contradiction is what pins the answer down.

Tools and results

(a) Invariants: three worked patterns

- **Geometric weighting.** Pebbles at lattice points, a pebble at (i,j) can be replaced by pebbles at $(i+1,j)$ and $(i,j+1)$. Give a pebble at (i,j) weight $2^{-(i+j)}$: each move preserves total weight. Total weight of the whole first quadrant is $\left(\sum_i 2^{-i}\right)^2 = 4$, and the weight of the points with $a+b \le 3$ is $1+2\cdot\frac{1}{2}+3\cdot\frac{1}{4}+4\cdot\frac{1}{8}=\frac{13}{4}$, so if no pebble sat on $a+b\le 3$ the total weight would be at most $\frac{3}{4}<1$ — but it is always exactly 1 (Indian TST 2004).
- **Congruence weighting.** n markers, move deletes a non-extreme white marker and flips its two nearest neighbours. Two invariants: the number of black markers is always even; and labelling a white marker with t black markers to its left by $(-1)^t$, the sum S of labels is invariant mod 3 (check the four local patterns WWW, WWB, BWB, BWW). At the end (two markers) $S\in\{0,2\}$ while initially $S=n$, so $n\equiv 0$ or $2 \pmod 3$ — i.e. the state with two markers is reachable *iff* $3\nmid(n-1)$ (ISL 2005 C5).
- **Parity of a composite quantity.** $m\times n$ board of markers, move deletes a black marker and flips its neighbours. If W counts white markers and E counts edges of the adjacency graph, a move deleting a black vertex of degree s with w white neighbours changes W by $s-2w$ and E by $-s$, so $W+E$ changes by $-2w$: **the parity of $W+E$ is invariant**. Initially $W+E=3mn-m-n-1\equiv(m-1)(n-1) \pmod 2$, so removing everything requires at least one of m,n odd (ISL 1998 C7).

(b) Good and bad objects

Call an object *good* if it has the property you want at the end. Two-step argument: (i) good objects stay good; (ii) as long as some object is bad, some move converts a bad object into a good one. Then

the number of good objects is a monovariant that strictly increases to the maximum, and the process stops. *Flagship:* $2n+1$ lamps in a circle, a lamp is good if it agrees with at least one neighbour; the odd number of lamps guarantees two adjacent equal lamps to start, and any bad lamp adjacent to a good lamp must flip and become good (based on Canada 1994).

(c) Bounding the number of steps

Do not bound the whole process; bound one *object's* contribution, then sum.

- **Per-object bound.** USAMO 2010-2 (students in a circle of distinct heights; switch places allowed when a student stands behind someone at least two height-ranks shorter). Let s_k be the number of times student k switches with a shorter student. Counting the students of height $< h_{k-1}$ between student k and student $k-1$ gives $s_k - s_{k-1} \le k-2$; with $s_1 = s_2 = 0$ this yields $s_k \le \binom{k-1}{2}$, and $\sum_k s_k \le \binom{n}{3}$ (using $\sum_k \binom{k-1}{2} = \binom{n}{3}$).
- **Recursive per-object bound.** Bookshelf of n volumes; a move takes a volume that is too far right and inserts it in its own position. Let $f(k)$ be the number of times volume k is chosen: $f(k) \le 1 + \sum_{i<k} f(i)$, and $f(1) = 1$, so $f(k) \le 2^{k-1}$ and the total is $< 2^n$ (based on Canada 2012 / ISL 1994 C4). Note the sum $\sum_{i<k} f(i)$: only smaller-labelled volumes can displace k.
- **Structure-driven bounds.** Belarus 2001 (a beetle on a $(2n-1)\times(2n-1)$ board of arrows, each arrow rotating 90° clockwise after the beetle leaves): split the board into boundary and interior, and corners from non-corners. A corner can be visited at most twice and a non-corner boundary square at most three times (a fourth visit would send the beetle off the board), and the interior is a $(2n-1)\times(2n-1)$ board to which induction applies.

(d) Problem alteration — changing the rules without changing the answer

Three instances, in increasing sophistication:

- **Ants on a stick.** Ants bounce off each other. Replace "bounce" by "pass through"; the multiset of positions is identical at every time, only the *labels* are permuted. Conclusion is immediate.
- **Reversal process (Russia 1993).** If the first entry is k, reverse the first k entries. Induct on n: once n appears first, it goes last and stays there for ever. If n never appears first, swapping the last entry j with n has no effect on the process at all. *"If a configuration is unreachable, permute the labels to make it so."*
- **Ants on a checkerboard (ISL 2011 C5).** When two ants moving in opposite directions meet they both turn 90° clockwise. Alter the rules twice: first declare that a rightward ant meeting a leftward ant becomes an up/down pair, so every ant is either "UR" or "DL" (a pass-through relabelling again); then, in a second alteration, declare every ant "UL" or "DR". The first alteration bounds collisions at time t by $t+1 \le x+y \le 2m-t-1$; the second bounds them by $|x-y| \le m-t-1$. Adding, a UR ant at its last collision has $x \ge t+1-\frac{m}{2}$, $y \ge t+1-\frac{m}{2}$, so the remaining straight run costs at most $m - \min\{x,y\} \le \frac{m}{2} - 1 + t$ and the total lifetime is at most $\frac{3m}{2} - 1$, which is attained by two ants.

(e) The terminal-state pattern: take the earliest bad configuration

Chip-firing (ISL 1994 C5). 1994 girls in a circle, one initially holds n coins, a move sends one coin from each girl with ≥ 2 coins to each neighbour. *(a) $n < 1994$ terminates:* suppose not; then every girl passes infinitely often. Having fixed the *first* coin ever passed across each of the 1994 gaps, we may assume that coin stays in that gap for ever (whenever one of the two neighbours passes a coin, we choose it to be that one). Every coin ends up permanently stuck in some gap; with fewer than 1994 coins some gap has no coin, contradicting infinity. *(b) $n = 1994$ never terminates:* weight a coin held by girl i by i; each move preserves the total weight mod1994 (it changes by 0 or ± 1994). The initial weight 1994 is $\equiv 0$, but the terminal configuration $1 + 2 + \cdots + 1994 = \frac{1994 \cdot 1995}{2}$ is not divisible by 1994.

Unique terminal state (ISL 2001 C7). Pebbles in columns; a pebble moves right when its column has at least two more pebbles than the next. Experiment with small n to conjecture that the final heights differ by at most one and take each value at most once except one value twice (for $n = 17$: $(5, 4, 3, 2, 2, 1)$). Prove it by taking the *earliest* configuration having two "bad" (equal) adjacent indices and examining the move just before it — that move must have come from one of the two bad columns, so the earlier configuration also had two bad indices. Contradiction.

Landmark problems from Ch. 3

Problem	Idea
Pebbles on a quadrant (Indian TST 2004)	Weight $2^{-(i+j)}$; total weight is invariant; a region of weight $< \frac{3}{4}$ cannot hold weight 1.
n markers (ISL 2005 C5)	Two invariants: parity of black markers, and the mod-3 label sum; answer: $3 \nmid (n-1)$.
$m \times n$ markers (ISL 1998 C7)	Parity of $W + E$; answer: at least one of m, n odd.
$2n + 1$ lamps (Canada 1994)	Good/bad monotone count.
Students switching (USAMO 2010-2)	Per-student recursive bound $s_k - s_{k-1} \le k - 2$; total $\binom{n}{3}$.
Bookshelf (Canada 2012)	Per-book bound $f(k) \le 2^{k-1}$; total $< 2^n$.
Beetle on an arrow board (Belarus 2001)	Boundary/interior and corner/non-corner visit counts; induction on the interior.
Sign chart $n \times n$ (Russia 2010)	Induction peeling off one row and column; case analysis on which line was operated.
Ants on a stick	Alter the rules: pass through instead of bounce.
Reversal process (Russia 1993)	Alter the rules: swap the last entry with n.
Ants on a checkerboard (ISL 2011 C5)	Alter twice; the two bounds $x + y$ and $\lvert x - y \rvert$ squeeze the last collision.
1994 girls, chip firing (ISL 1994 C5)	"Stuck coin" argument for termination; weight mod1994 for non-termination.
Unique pebble configuration (ISL 2001 C7)	Earliest bad configuration; answer $(5, 4, 3, 2, 2, 1)$ for $n = 17$.

Key takeaways

- **Run the process on small cases before proving anything.** The final configuration of the pebble problem, the threshold in the chip-firing game, and the constant $\frac{3m}{2} - 1$ in the ant problem were all *conjectured* first.
- **Two invariants are better than one.** Exactly one invariant rarely decides a problem; the marker problem needs the black-parity invariant *and* the mod-3 label invariant.
- **Rule alteration is legal if the change is a relabelling.** Passing through instead of bouncing, or swapping two labels, changes who-is-who but not the answer to a question about positions or counts.
- **Restrict the process to make it analysable.** In the ant problem, the ants are forced into two direction classes — a restriction that only helps the proof.
- **To pin down a terminal state, look at the last move before the *first* time something bad happens.** Extremality + the move set is the standard uniqueness proof.
- **Bound each object's contribution, then sum.** This is how all polynomial step bounds are obtained.
- **If the process does not terminate, look for a conserved quantity whose value at**

the end is impossible. That is the whole content of part (b) of the chip-firing problem.

Where it usually goes wrong

- **A "stuck coin" style argument needs the choice to be consistent.** You must be allowed to keep choosing the *same* coin at every future pass; the book's phrasing ("we may assume the coin passed was c_i") is doing real work.
- **Monovariants that are bounded but not well-ordered.** A decreasing sequence of positive *reals* need not terminate; make sure the quantity is a non-negative integer or that it decreases by a fixed amount.
- **Assuming the final configuration is unique because the problem's "obvious" answer is unique.** Uniqueness is a theorem, not a definition — and the problem is usually exactly about it.
- **Applying a rule alteration that changes a count.** "Pass through" preserves positions but not identities; if the question is about *which* ant falls off, the alteration is illegal.

4. Existence

Snapshot

Ch. 4 is the pivot of the book. **Ch. 1–Ch. 3** proved existence by *construction*; **Ch. 4** proves it without one, and also proves *non-existence*. The tools are new and mostly indirect: the extremal principle (and its child, infinite descent), the *optimal assumption* ("take the largest object with the property and use maximality"), invariants (again), the *hostile neighbours* principle, divide and conquer, and discrete continuity. The unifying theme is proof by contradiction.

The essential idea

The optimal assumption. To prove that a set of size $\geq X$ satisfying property P exists, do *not* construct one. Let S be a *largest* set satisfying P, and then *use* maximality: for every $x \notin S$ something must break if you add x — and that "something" is a machine that manufactures structure, usually an injection.

Hostile neighbours. If n objects in a sequence (or on a circle) are of two types and both types occur, then some adjacent pair has opposite types. This one sentence proves Redei's theorem, the $n \times n$ board problem, and many others. It is the discrete cousin of the intermediate value theorem — which is exactly the next idea.

Discrete continuity. If a sequence of integers satisfies $|a_{i+1} - a_i| \leq 1$, it attains every integer between a_1 and a_n. In particular it crosses 0.

Tools and results

(a) Induction, top-down

ISL 1985: $3n+2$ points around a circle marked ± 1, at most n of them -1; prove a *good* point exists (all partial sums in both directions are positive). Note $1985 = 3 \cdot 661 + 2$. The induction step *deletes* a block $1, -1, \ldots, -1, 1$ (a maximal run of -1s together with its two bordering $+1$s), reducing $3(k+1)+2$ points with $\leq k+1$ minus-ones to $3k+2$ points with $\leq k$ minus-ones. Top-down works here because *you may choose* which points to delete; bottom-up would lose generality.

(b) The extremal principle and infinite descent

- **Minimum prefix sum.** France 1997: labels on a 1997-gon with total 1; find a starting vertex making all partial sums positive. Interpret the labels as altitude changes and start at the *lowest* point. Formally, let b_k be the prefix sums and take k with $b_k = \min_j b_j$ (largest such k); starting at

V_{k+1} works, because a negative partial sum would contradict minimality. This is the "cycle lemma" in disguise and is worth memorising.

- **Infinite descent.** If every object with property P can be used to produce a *smaller* object with P, then in a finite set P is impossible. *Flagship:* Indian TST 2003 — a partition $\{1,\dots,3n\} = A \sqcup B \sqcup C$ with $|A| = |B| = |C| = n$ must contain $x \in A$, $y \in B$, $z \in C$ with one the sum of the other two. Assume not, put $1 \in A$, let k be the least integer not in A (so $k \in B$). Then no C-element and A-element differ by k, and no B-element and C-element differ by less than k. Taking $m, m-1 \in C$ produces $m-k, m-k-1 \in C$ — a strictly smaller such pair. Descent is impossible in a finite set, so $m \in C \Rightarrow m-1 \in A$, giving an *injection* $C \to A$; since $|A| = |C| = n$ it is a bijection, which forces $2 \in C$, contradicting $k \in B$.
- **Descent on a well-chosen measurement.** ELMO Shortlist 2012 (grid colouring where every square has exactly two same-coloured neighbours, one of m, n odd): find an odd-length monochromatic *good chain* in the top row, then argue the row below forces a shorter good chain, and descend until a chain of length 1 — impossible.

(c) The optimal assumption in action

- **Simple version.** If every vertex of G has degree $\geq \delta$, then G has a path on at least $\delta + 1$ vertices: take a *longest* path and look at its last vertex — all its δ neighbours are on the path.
- **Injection from maximality.** Italy TST 1999: sets $A_1, \dots, A_m$ of size 3 with $|A_i \cap A_j| \leq 1$; show a subset of X (of size n) with at least $\lfloor \sqrt{2n} \rfloor$ elements containing no A_i. Take the largest such S, $|S| = k$. Every $x \notin S$ must complete some A_i with a *pair* of elements of S — else $S \cup \{x\}$ still works. The map $x \mapsto \{y, z\}$ is injective (two x's mapping to one pair would give two A_i's meeting in two points). Hence $n - k \leq \binom{k}{2}$, i.e. $k^2 + k \geq 2n$.
- **Maximum cut.** Every graph has a partition $V = V_1 \sqcup V_2$ in which every vertex has at least as many neighbours across as inside: take the partition *maximising the number of crossing edges*; a violating vertex could be moved to increase it. (The book notes that the *algorithmic* max-cut problem is hard — existence is easy, computation is not.)

(d) Invariants as an existence/classification tool

Italy TST 1995: an 8×8 board tiled by 21 trominoes leaves exactly one square uncovered. Colour the board periodically with 3 colours so that *every* tromino covers one square of each colour; then the covered multiset is 21 of each colour, so the uncovered square must be the colour that occurs 22 times. Repeat with the mirror colouring. A square must have the same colour in both — this leaves only 4 candidates, and each of them is constructible.

(e) Hostile neighbours and discrete continuity

- **Redei's theorem.** Every tournament has a directed Hamiltonian path. Induct: delete a vertex v, keep the path $v_1 \to \cdots \to v_{n-1}$. If $v \to v_{n-1}$ or $v_1 \to v$ we are done; otherwise there is an index k with $v_k \to v$ and $v \to v_{k+1}$ — *because the sequence of comparisons* $v_1 \to v,\ v_2 \to v, \dots$ *is of two types and changes type*, and one splices v between v_k and v_{k+1}.
- **The n^2 board (ISL 1988).** Numbers $1, \dots, n^2$ on an $n \times n$ board; some adjacent pair differs by at least n. Assume all differences $\leq n-1$; define $S_k = \{1, \dots, k\}$, $N_k = \{k+1, \dots, k+n-1\}$, $T_k = \{k+n, \dots, n^2\}$. Some row and some column avoid N_k; their union is a *cross*, and the cross is monochromatic in the $\{S_k, T_k\}$ classification. For $k = 1$ the cross must lie in T_k; near the top it must lie in S_k; by discrete continuity there is an intermediate j where the two crosses (for N_j and N_{j+1}) have conflicting types — and they intersect in two squares that would have to be in both S_{j+1} and T_j.
- **Balancing lines (USAMO 2005 P5).** $2n$ points, n blue and n red, no three collinear; at least two *balancing* lines exist. If the hull has both colours, two adjacent hull points of different colours give two balancing lines. If the hull is monochromatic (say blue), take a hull point P, rotate a line about it, and let $f(i) = b_i - r_i$ be the blue-minus-red count before the i-th red point. Then

$f(1) \geq 0$ and $f(n) \leq 0$, and f decreases by at most 1 per step: discrete continuity gives $f(i) = 0$, i.e. PR_i is balancing. Repeating for each hull point gives at least three.

(f) Divide and conquer and a counting bound

ISL 2002 C2 (L-trominoes covering the black squares of an odd $n \times n$ board with black corners): the key is to *select a subset of squares no tile can cover twice*. The black squares on odd rows form such a set, and there are $(k+1)^2$ of them for $n = 2k+1$; each L-tromino covers at most one. Hence at least $(k+1)^2$ tiles, and $(k+1)^2$ tiles are also sufficient for $n \geq 7$; small n fail by area.

(g) Miscellaneous, but instructive

- **Pigeonhole then extremal.** Romania 2001 (three schools of 200; a set E of 300 with a distinctness condition): some school contains at most 100 of E, so ≥ 200 members of E lie outside it; between 1 and 200 distinct values of "number of friends in S_1" force one student to have 200. Then close the triangle.
- **Double counting at the existence level.** ISL 1988 (even n; $n+1$ sets of n elements, pairwise meeting in exactly one, every element in ≥ 2 sets): show first that every element lies in *exactly* two sets, then count zeros: $k(2k+1)/2$ must be an integer, so $4 \mid n$.
- **Self-descriptive sequences (ISL 2001 C5).** x_j equals the number of occurrences of j in the sequence. Let m be the number of non-zero terms; the non-zero terms sum to m, so exactly $m-2$ of them are 1 and one is 2; this bounds $m \leq 4$ and gives the complete list $(2,0,2,0)$, $(1,2,1,0)$, $(2,1,2,0,0)$ and $(k,2,1,0,\ldots,0,1,0,0,0)$.

Landmark problems from Ch. 4

Problem	Idea
$3n+2$ points, good point (ISL 1985)	Top-down induction: delete a maximal run of -1s and its bordering $+1$s.
Partition of $\mathbb{N}$ (ISL 1990)	Assume no part is good, then inductively prove that the tail union contains arbitrarily long runs of consecutive integers — contradiction at $s = r$.
Labels summing to 1 (France 1997)	Start at the lowest point; minimum prefix sum.
Partition of $\{1,\ldots,3n\}$ (Indian TST 2003)	Least integer outside A; infinite descent on consecutive C-pairs; the descent becomes a bijection $C \to A$.
Grid colouring (ELMO SL 2012)	Odd monochromatic chains; descent to a chain of length 1.
Path in a graph of min degree δ	Longest path, look at the endpoint.
Avoid a family of triples (Italy TST 1999)	Largest avoiding set; injective map $X \setminus S \to$ pairs; $k^2 + k \geq 2n$.
Max cut	Partition maximising crossing edges; move a violating vertex.
8×8 board, 21 trominoes (Italy TST 1995)	Periodic 3-colouring plus its mirror; only 4 squares can be uncovered.
Tournament Hamiltonian path (Redei)	Induction plus a sign change in the comparison sequence.
n^2 board, adjacent difference $\geq n$ (ISL 1988)	$S_k/N_k/T_k$ and the *cross* argument; two crosses intersect in a forced contradiction.
Balancing lines (USAMO 2005 P5)	$f(i) = b_i - r_i$ crosses 0 with steps of size ≤ 1.

Problem	Idea
L-trominoes (ISL 2002 C2)	Choose a set of black squares no tile can cover twice; answer $(k+1)^2$ for $n \geq 7$.
Three schools (Romania 2001)	Pigeonhole to find a student friendly with an entire school; close the triangle.
$n+1$ sets, pairwise one common element (ISL 1988)	Every element in exactly two sets; $k(2k+1)/2 \in \mathbb{Z}$ forces $4 \mid n$.
Self-descriptive sequences (ISL 2001 C5)	m non-zero terms sum to m; four solutions.

Key takeaways

- **When you cannot construct, extremise.** Take the largest/smallest/ longest/leftmost object with the property and squeeze it.
- **Maximality is a factory for injections.** "Adding x would break P" means x has a *witness* inside S; if witnesses are unique, you get an injection, hence an inequality.
- **Infinite descent proves non-existence.** If property P regenerates a smaller instance of itself in a finite set, P is empty — and often the descent step itself hands you the injective map you need.
- **Hostile neighbours: two types in a line $\Rightarrow$ a boundary.** Formalise "there is a place where the type changes".
- **Discrete continuity: steps of size ≤ 1 cross everything in between.** Look for an integer-valued sequence that starts non-negative and ends non-positive.
- **A good choice of Q to count beats a good construction.** When existence is at stake, count.
- **Necessary from invariants, sufficient from algorithms.** Most "find all n" problems are one invariant plus one construction; do not stop after the invariant.

Where it usually goes wrong

- **Using maximality without saying what breaks.** "Take the largest S" is only useful once you have written the sentence "then $S \cup \{x\}$ would also have the property, contradicting maximality".
- **Descent on a quantity that is not well-ordered** (e.g. on a positive real). The descent must land in a finite set.
- **Assuming the injection is a bijection.** Indian TST 2003 needs $|A| = |C|$; that hypothesis is not decoration.
- **Extremal principle on an infinite set.** "The smallest element" need not exist in $\mathbb{Z}$ or $\mathbb{Q}$; the technique is for finite sets (or well-ordered ones).
- **In discrete continuity, forgetting the step-size hypothesis.** The conclusion fails for arbitrary integer sequences: you must verify $|a_{i+1} - a_i| \leq 1$.
- **Confusing the two directions of a classification.** "$f(i) = 0$ iff PR_i is balancing" must be proved both ways before you use it.

Part II Compete and Count

5. Combinatorial Games

Snapshot

A combinatorial game has (i) no chance, (ii) perfect information, (iii) alternating moves by two players, and (iv) no cheating. A typical problem gives a starting position, the legal moves, and the winning condition, and asks *who wins* — or, harder, *what the winning strategy is*.

The chapter develops four families of ideas:

1. **Symmetry, pairing and copying** — reply to the opponent's move with a mirror or a partner move.
2. **Parity-based pairing** — often you only need to *prove a pairing exists*, and an even number of objects is enough.
3. **Positional analysis** — classify every position as P or N and always move to a P-position.
4. **Maker–breaker invariants** — when the two players have different goals, positional analysis must be replaced by a maintained structural invariant.

The essential idea

The fundamental theorem of finite games. A finite game with no ties and no chance cannot end in a draw, so one of the players has a winning strategy. This is used constantly, and it has a powerful corollary used in **Ch. 4**-style proofs:

If you can show that a player can always "stay alive" (always has a legal reply), then that player wins.

Since the game is finite, the opponent must eventually be the one who cannot move.

Positional analysis in one paragraph. A position is a **P-position** if the player who has *just moved* can force a win, and an **N-position** if the player *to move* can force a win. Winning strategy = always hand the opponent a P-position. Compute P-positions by working backwards from the terminal positions: n is P iff every legal move from n leads to an N-position; n is N iff some move leads to a P-position.

Tools and results

(a) Symmetry, mirroring and copying

- **Mirror about the centre.** Coins on a finite table: the first player places a coin at the centre, then reflects every one of the opponent's moves in the centre. The configuration is symmetric after each of the first player's turns, so a legal move for the opponent always has a legal mirror. Finite area forces termination; the first player wins.
- **Copy a structure, not a position.** N is a product of $k \geq 3$ distinct primes; players write composite divisors of N with no two coprime and no one dividing another. The first player writes pq for two primes and thereafter answers qm with qn (or symmetrically). Every subsequent number must be of the form pm or qm, so the reply always exists. *The copying strategy is a bijection on the move set.*
- **Pairing to control where a *maximum* lands.** USAMO 2004-4: on a 6×6 grid, Alice and Bob write rationals; the maximum of each row is coloured black, and Alice wins if the black squares connect top to bottom. Bob pre-selects a set of squares X (three per row, staggered so that no black path exists) and pairs each square of X in a row with a square outside X. When Alice plays in X Bob plays a *smaller* number in the partner; when Alice plays outside X Bob plays a *larger* number there. After every Bob move the row maximum is inside X.
- **Pairing to block all winning lines.** In the 5×5 tic-tac-toe game there are 12 winning lines (5 rows, 5 columns, 2 diagonals), and 25 squares, so 12 pairs cover 24 squares and each winning line contains exactly one pair. Bob answers the partner of Alice's square. Since every winning line contains a pair, Bob occupies at least one square of every line. *Recipe: count the lines, halve the board, cover each line exactly once.*
- **Domino pairing to bound a maximum.** ISL 1994 C1: Alice writes 1, Bob writes 0 on a 5×5

board, Alice's score is the largest sum in a 3×3 sub-board. Bob tiles 10 dominoes leaving the bottom row; every 3×3 square contains at least three complete dominoes, hence at least three 0s, so $S \leq 6$. The bound is sharp.

(b) Parity-based pairing — existence is enough

If a set has an even number of objects, it can be partitioned into pairs — and sometimes that is all the structure you need.

- **Italy TST 2009 (permutation game).** Alice writes a permutation of $1, \ldots, n$; each subsequent move writes a *new* sequence that either is a permutation of the previous one or deletes one entry from it. For $n = 2$ Bob wins by deleting. Inductively, Bob's aim is to force Alice to be the first to delete: since $(k+1)!$ is even, after Alice writes a $(k+1)$-permutation there is always an unused permutation for Bob to write. Then the game reduces to k entries and Bob wins by induction. (Explicit alternative: Bob writes the reverse of Alice's sequence.)
- **USAMO 1999-5 (the Y2K game).** On a 1×2000 strip players write S or O; the first to produce consecutive *SOS* wins. Call a square *bad* if playing there lets the opponent complete SOS next move. *Claim:* a square is bad iff it lies in a block $S__S$. Bad squares therefore come in pairs, so at every one of Bob's turns (after an odd number of moves) there is an even number of bad squares and an odd number of empty squares — hence at least one legal, non-losing move. Bob sets up a block $S__S$; since the game is finite and Bob never loses, Bob wins.

(c) Positional analysis

Worked template. *Ex. 8:* players add 1, 2 or 3 to a running total, starting from Alice's 1, 2 or 3; the player who says 100 wins. Working backwards: 100 is a P-position, so $97, 98, 99$ are N; 96 is P; ... by induction every multiple of 4 is a P-position. Bob always answers with the next multiple of 4. *Generalisation:* with moves $1, \ldots, k$ and target n, the P-positions are the numbers $\equiv r \pmod{k+1}$ where $n \equiv r$; if $r = 0$ then 0 is P and the second player wins, otherwise the first player wins by saying r.

Other shapes:

- **Two-dimensional P-positions.** Lithuania 2010: a stone on an $m \times n$ board at the lower-left, moves go up or right any number of squares, and the player who moves it to the upper-right wins. After reflecting the board, a P-position is a square with equal coordinates. If $m \neq n$ the start is N and Alice wins by moving to a diagonal square; if $m = n$ the start is P and Bob wins. *The P-positions form a diagonal; the winning strategy is to stay on it.*
- **Partial classification is enough.** St Petersburg 2001: from n, replace it by $n-1$ or $\lfloor n/2 \rfloor$; the player who writes 1 wins. Every even number is an N-position: for $2k$, either k is a P-position and Alice writes k and wins, or k is an N-position and Alice writes $2k-1$, after which Bob must write $2k-2$ or k, both N-positions. *We never determine whether odd numbers are P or N — and we do not need to.* The problem asks *who* wins, not *what* the strategy is; a proof that a winning strategy exists suffices.
- **Existential positional analysis.** Russia 2011: $N > n^2$ stones, a move takes k stones where $k < n$ or $n \mid k$. Suppose Bob wins. Alice removes kn; Bob responds $f(k)$ with $1 \leq f(k) \leq n-1$, so by pigeonhole $f(k) = f(j)$ for some $k \neq j$. Then Alice can reproduce the position after Bob's response in the other case, reaching a P-position, contradiction. *The extremal principle is replaced by a pigeonhole on responses.*
- **Recursion down to a smaller game.** ISL 2004 C5: Alice writes 1, Bob writes 2; then from k write $k+1$ or $2k$ (never exceeding N); the writer of N wins. Odd N: Alice always writes odd numbers, so Bob can never write N. For even N: every even number $> N/2$ is a P-position (nobody can double any more, so the parity of the parity-alternating run decides). Consequently k is a P-position when $N \in \{4k, 4k+2\}$, and one shows that a win for $N = k$ transfers to $N = 4k$ and $N = 4k+2$. Hence the answer is read off from base 4: **Alice wins iff N has an odd digit in base 4; otherwise Bob wins.**

(d) Maker–breaker: invariants instead of P/N

When the players have opposite objectives, P- and N-positions no longer make sense. The substitute is a *structural invariant that one player can maintain.*

ISL 2009 C5 (Cinderella and the Stepmother). Five buckets of capacity 2 at the vertices of a pentagon. Each round the Stepmother adds one litre anywhere; Cinderella empties two adjacent buckets. The Stepmother wants an overflow. Call a configuration *good* if $B_i + B_{i+2} \le 1$ for every i; in a good position no two buckets at distance two can both exceed 1, so no overflow can be forced immediately. Cinderella's invariant: *from a good configuration, she can always return to a good configuration.* Indeed if $B_4 = B_5 = 0$ in a good configuration, then after the Stepmother's one litre the total in $\{B_1, B_3, B_4, B_5\}$ is at most 2, so either $B_5 + B_3 \le 1$ or $B_4 + B_1 \le 1$; emptying B_1, B_2 in the first case restores goodness. Since the empty board is good, Cinderella never overflows.

Landmark problems from Ch. 5

Problem	Method
Coins on a table	Centre first, then mirror in the centre; "always stay alive".
Composite divisors of N (St Petersburg 1997)	Write pq, then copy: $pm \mapsto pn$, $qm \mapsto qn$.
6×6 grid (USAMO 2004-4)	Pair each row's "forbidden" squares with allowed ones; control where each row maximum falls.
5×5 tic-tac-toe	12 winning lines, 12 pairs covering 24 squares; each line meets exactly one pair.
5×5 board, sums (ISL 1994 C1)	Domino tiling leaving a row; every 3×3 has ≥ 3 full dominoes; answer 6.
Permutation game (Italy TST 2009)	$(k+1)!$ even; Bob forces Alice to delete first, then inducts.
Y2K game (USAMO 1999-5)	Bad squares are exactly blocks $S__S$, hence come in pairs; Bob never gets stuck.
Adding $1, 2, 3$ to 100	Multiples of 4 are P-positions; generalisation mod $k+1$.
Stone on an $m \times n$ board (Lithuania 2010)	P-positions are the diagonal (k, k); Alice wins iff $m \ne n$.
Halving game (St Petersburg 2001)	All even numbers are N-positions — a partial classification suffices.
$N > n^2$ stones (Russia 2011)	Contradiction plus pigeonhole on the opponent's replies.
$+1$ or $\times 2$ to N (ISL 2004 C5)	P-positions >N/2; recursion $N \to \lfloor N/4 \rfloor$; base-4 answer.
Cinderella (ISL 2009 C5)	Maintain the invariant $B_i + B_{i+2} \le 1$ after each of her turns.

Key takeaways

- **Before analysing, ask whether the game is finite.** If yes, someone wins; if you can always move, you win.
- **Symmetry is the cheapest strategy.** If you can make the position symmetric after your move, you can always answer your opponent.
- **Pairing beats cleverness.** Many "Bob wins" problems are just a partition of the board into pairs, one per winning line.
- **You may only need to prove a pairing exists.** An even number of objects is enough (the permutation game).
- **Find the P-positions by working backwards, then look for a pattern.** Multiples of $k+1$; equal coordinates; powers of two.

- **You do not always need a full classification.** If the problem asks only *who* wins for one starting value, a partial classification plus "existence of a strategy" suffices.
- **For asymmetric games, look for an invariant one player can maintain.** Cinderella's $B_i + B_{i+2} \le 1$ plays the role of a P-position.
- **Recursion in games: a win for size k often upgrades to size $4k$ or $4k+2$.** Look to base-4 (or base-b) representations for the closed form.

Where it usually goes wrong

- **Applying P/N language to maker–breaker games.** With different objectives there is no "previous player wins".
- **Using positional analysis on a game with cycles.** The recursion "n is P iff all moves lead to N" presumes the game ends; a game with cycles may never terminate.
- **Mirroring the wrong symmetry.** The mirror must map legal moves to legal moves *and* leave the centre fixed; mirroring about the wrong axis breaks the argument.
- **Assuming the first player wins in every symmetric game.** Symmetry arguments often favour the *second* player (the copycat); check who moves first after the symmetrising move.
- **Pairing that does not cover every winning line.** In the tic-tac-toe problem, the pairing is engineered so that each of the 12 lines contains a pair; an arbitrary pairing of 24 squares does not work.

6. Counting in Two Ways

Snapshot

Ch. 6 is the engine room of the book. The idea could not be simpler — compute one quantity Q in two different ways and equate (or compare) the two answers — but it is the technique that converts a combinatorial hypothesis into algebra, and it underlies most of **Ch. 7** and **Ch. 8**.

The reason it is so powerful is that the two countings usually *use different hypotheses*. One way uses what the problem says about sets; the other uses what it says about elements. When the problem gives you bounds rather than equalities, the two countings give an inequality, and that inequality is frequently the whole problem.

The essential idea

Choose Q, then choose the two ways. The two workhorses:

Count 1-entries of an incidence matrix	one way by rows ($\sum_i \lvert A_i\rvert$), the other by columns ($\sum_j d_j$). Equating gives the incidence identity; bounding one side gives inequalities.
Count pairs/triples with a side condition	(set, element), (set, set, element) for *intersection* information, (element, element, set) for *co-occurrence* information.

Read the problem's quantifiers. The words *at most* and *at least* tell you which counting direction yields a bound, and often tell you exactly what to count. This is the single most useful heuristic in the chapter.

Tools and results

(a) Incidence matrices and the master identity

For subsets $A_1, \dots, A_m$ of $\{1, \dots, n\}$ with element j lying in d_j of them,

$$\sum_{i=1}^{m} |A_i| \;=\; \sum_{j=1}^{n} d_j.$$

Both sides count the 1s of the incidence matrix (rows = sets, columns = elements), one by rows and one by columns. **Variations:**

- count triples (set, set, element) — pairs of 1s in the same *column*; useful when intersection sizes are given;
- count triples (element, element, set) — pairs of 1s in the same *row*; useful when co-occurrence counts are given.

Balanced block designs. A (v, k, λ) design (every block has k elements, every pair of elements lies in exactly λ blocks) satisfies

$$r(k-1) = \lambda(v-1) \qquad \text{and} \qquad bk = vr,$$

where b is the number of blocks and r the number of blocks through a point. The first comes from counting (element, element, block) triples with the two elements distinct and one of them fixed; the second from counting (element, block) incidences. *Note the payoff: the first identity proves that every element lies in the same number of blocks — not obvious from the definition.*

(b) Getting inequalities out of double counting

- **USA TST 2005.** $2n$ sets, each of size m, pairwise sharing at most one element, each element of $S_m = \{1, \dots, mn\}$ in exactly two of them. Let S be the number of triples (x, A_i, A_j) with $x \in A_i \cap A_j$. Fixing x: exactly one pair, so $S = mn$. Fixing the pair: at most one x, so $S \le \binom{2n}{2}$. Hence $mn \le \binom{2n}{2}$, i.e. $m \le 2n-1$. Sharp: $2n$ lines in general position.
- **IMO 1998 P2.** a contestants, b odd judges, any two judges agree on at most k contestants. Count triples (two judges, one contestant on whom they agree):

$$T \le k\binom{b}{2} \quad \text{and} \quad T \ge a\left[\binom{p}{2} + \binom{b-p}{2}\right] \ge a \cdot \frac{(b-1)^2}{4},$$

  where p is the number of "pass" ratings for a fixed contestant (convexity, and b odd makes the minimum exact). Hence $k/a \ge (b-1)/(2b)$.
- **IMO 1987.** n points, no three collinear, each point at distance equal to at least k other points. Each point is the centre of a circle through $\ge k$ points; counting pairs of points on circles gives $n\binom{k}{2} - \binom{n}{2} \le \binom{n}{2}$, hence $k < \frac{1}{2} + \sqrt{2n}$. *The overcount is bounded by the number of circle-pairs, since two circles meet in at most two points.*
- **IMO Shortlist 2004 C1.** 10001 students, clubs, societies, with (i) each pair of students in exactly one club, (ii) each student and society determine exactly one club, (iii) a club with $2m+1$ students lies in exactly m societies. Count triples (student, club, society) by (student, society, club): $10001k$. By (club, society, student): $\sum_C \frac{|C|-1}{2}$, which equals the number of pairs of students. Hence $10001k = \binom{10001}{2}$ and $k = 5000$.
- **Geometric inequality (Iran 2010).** n points, no three collinear: at most $\frac{2}{3}n(n-1)$ triangles of area 1. Count pairs (edge, triangle of area 1): each triangle has 3 edges so the count is $3k$; for a fixed edge at most 4 points can form a unit-area triangle with it (they lie on two lines parallel to the edge at fixed distance, and no three points are collinear). Hence $3k \le 4\binom{n}{2}$.

(c) Counting with graphs — the identities you should know

Let G have degrees $d_1, \dots, d_n$ and k edges. Then:

$$\textbf{(1)}\ \sum_i d_i = 2k, \qquad \textbf{(2)}\ \sum_i d_i^2 \ \ge\ \frac{\left(\sum_i d_i\right)^2}{n} = \frac{4k^2}{n}\ \text{(Cauchy–Schwarz)},$$

$$\textbf{(3)}\ \sum_i \binom{d_i}{2} \ \ge\ \frac{2k^2}{n} - k, \qquad \textbf{(4)}\ \sum_i \sum_{j \sim i} d_j = \sum_i d_i^2$$

((4) because d_i appears d_i times on the left). For a tournament with w_i wins and ℓ_i losses,

$$\textbf{(5)}\ \sum_i w_i = \sum_i \ell_i = \binom{n}{2}, \qquad \textbf{(6)}\ \sum_i \binom{w_i}{2} = \sum_i \binom{\ell_i}{2}.$$

Identity (6) is proved by counting *non-cyclic* triples by their winner ($\sum \binom{w_i}{2}$) and by their loser ($\sum \binom{\ell_i}{2}$). The book's remark is a general habit worth adopting: *a sum of squares or a sum of binomials should be given a combinatorial interpretation* before you try to bound it.

- **APMO 1989.** Every graph with n vertices and k edges has at least $\frac{k(4k-n^2)}{3n}$ triangles. For an edge v_iv_j, the number of common neighbours is at least $\max\{0, d_i + d_j - n\}$ (inclusion–exclusion with $n-2$ other vertices); summing over edges with (3) and (4) and dividing by 3 gives the bound.
- **Corollary (Mantel).** A triangle-free graph has at most $\lfloor n^2/4 \rfloor$ edges, with equality only for balanced complete bipartite graphs. This is the $r = 2$ case of Turán's theorem (**Ch. 8**, exercises).
- **No 4-cycles (Indian TST 2001).** Count "V-shapes" $(u; \{v, w\})$ with v, w both adjacent to u. Fixing the outer pair gives at most one u (else a 4-cycle), so $T \le \binom{n}{2}$; fixing u gives $T = \sum_u \binom{d_u}{2} \ge \frac{2E^2}{n} - E$. Combine and solve the quadratic: $E \le \frac{n}{4}(1 + \sqrt{4n-3})$.
- **USAMO 1995.** In a graph with q edges and no triangles, count pairs (edge, vertex) where the edge contains the vertex or a friend of it; Cauchy–Schwarz gives $P = \sum_i d_i^2 \ge 4q^2/n$, and averaging produces a vertex whose foes span at most $q(1 - 4q/n^2)$ amicable pairs. *When you want to count objects* not *satisfying a property, count the complement and subtract.*
- **ISL 2010 C5 (no "bad company").** Four players form a bad company if one lost to the other three and those three form a cyclic triple. Assuming no bad company, prove $\sum w_i^3 \ge \sum \ell_i^3$. Write $\sum w_i^3 - \sum \ell_i^3$ using (6) as a comparison of $\sum \binom{w_i}{2}$ and $\sum \binom{\ell_i}{2}$ plus a comparison of $\sum \binom{w_i}{3}$ and $\sum \binom{\ell_i}{3}$, and interpret each as counting *chained quadruples* by winner/loser, and quadruples together with a player who won exactly 1 resp. exactly 2 of the three internal games. Quadruple types $(3,1,1,1)$, $(2,2,1,1)$, $(3,2,1,0)$ are allowed and $(2,2,2,0)$ is exactly the forbidden company; comparing type by type proves the inequality. *This is the book's showcase for "keep translating into combinatorial statements and let the translation tell you what to count".*

(d) Miscellaneous applications — three transferable tricks

- **Sum a family modulo something (IMO 2001 P4).** For odd $n > 1$, among $n!$ permutations a with $S(a) = \sum i\, c_{a_i}$, two have $n! \mid S(a) - S(b)$. Suppose all $S(a)$ were distinct mod $n!$; then they would be exactly $1, 2, \ldots, n!$, whose sum is $n!/2 \pmod{n!}$. But the coefficient of every c_i in $\sum_a S(a)$ is $(n-1)!(1 + \cdots + n) = n! \cdot \frac{n+1}{2} \equiv 0$, so $\sum_a S(a) \equiv 0$. Contradiction.
- **Weight the terms so the signs cooperate (ISL 2003 C4).** If A is the 0–1 matrix $a_{ij} = 1 \iff x_i + y_j \ge 0$ and B is any 0–1 matrix with the same row and column sums, then $A = B$. Set $S = \sum_{i,j}(x_i + y_j)(a_{ij} - b_{ij})$: the row/column-sum hypothesis makes $S = 0$, while each term satisfies $(x_i + y_j)(a_{ij} - b_{ij}) \ge 0$. So every term is 0, which forces $a_{ij} = b_{ij}$ wherever $x_i + y_j \ne 0$, and the equal-sum hypothesis handles the rest. *Moral: to prove many reals are zero, make a non-negative sum that is known to be zero — and then* weight *the terms so that the available hypotheses can be used.*
- **Sum over all rectangles (Indian TST 2010).** In a 10×10 array of positive reals with all row and column sums 1, some "diagonal pair" ratio is at least $1/50$. Interpret a rectangle's value as $a_{jl}a_{km}/(a_{jm}a_{kl})$; the sum S of the 45^2 rectangle values counts each product $a_{ij}a_{kl}$ (with $i \ne k$, $j \ne l$) — so $S = \frac{1}{2}\sum_{i,j} a_{ij}S_{ij}$ with $S_{ij} = 8 + a_{ij}$, and Cauchy–Schwarz gives $S \ge 4 \cdot 10 + \frac{1}{2} = 40.5$, contradicting $S < 45^2/50 = 40.5$.

Landmark problems from Ch. 6

Problem	What is counted, and how
Incidence matrix	1s, by rows and by columns — the master identity.
Block designs	(element, element, block) triples for $r(k-1) = \lambda(v-1)$; (element, block) pairs for $bk = vr$.
USA TST 2005	Triples (x, A_i, A_j) with $x \in A_i \cap A_j$; $mn \le \binom{2n}{2}$.
IMO 1998 P2	(judge, judge, contestant) triples with agreement; convexity for the lower bound.
Iran 2010 (unit-area triangles)	(edge, triangle) pairs; at most 4 points per edge.
IMO 1987 (equidistant points)	Pairs of points on circles; overcount $\le$ number of circle pairs.
ISL 2004 C1 (clubs and societies)	(student, club, society) triples twice; $k = 5000$.
APMO 1989 (triangles)	(edge, triangle) pairs; inclusion–exclusion $d_i + d_j - n$.
Indian TST 2001 (no 4-cycles)	V-shapes; $T \le \binom{n}{2}$ and $T \ge \frac{2E^2}{n} - E$.
USAMO 1995 (friendly foes)	Pairs (edge, vertex) through friends; Cauchy–Schwarz.
ISL 2010 C5 (no bad company)	Chained quadruples and quadruple types; the forbidden type is $(2, 2, 2, 0)$.
IMO 2001 P4 (permutations)	Sum all $S(a)$ modulo $n!$; each coefficient vanishes.
ISL 2003 C4 (two matrices)	Weighted non-negative sum $\sum(x_i + y_j)(a_{ij} - b_{ij})$.
Indian TST 2010 (rectangles)	Sum rectangle values; reduce to Cauchy–Schwarz on the a_{ij}.

Key takeaways

- **Two countings, two hypotheses.** Choose Q so that each way of counting uses a *different* piece of the problem statement.
- **Quantifier ⇒ direction.** "At most" gives an upper bound on one count; "at least" gives a lower bound on the other. Combining them is the whole argument.
- **Divide by the multiplicity.** Each triangle is counted 3 times (once per edge), each unordered pair twice. Forgetting this is the most common error in the chapter.
- **Translate sums into combinatorial objects.** $\binom{d_i}{2}$ is a pair of edges at v_i; d_i^2 counts ordered pairs; $\binom{w_i}{2}$ is a pair beaten by i. Let the algebra suggest the object.
- **When a direct count is hard, count the complement.** USAMO 1995 asks for "few amicable pairs among foes"; the proof counts all pairs and subtracts.
- **Inclusion–exclusion with degrees.** Two neighbour-sets inside an n-vertex graph have intersection at least $d_i + d_j - n$; this "+ and −" move recurs in every extremal triangle count.
- **Weighted non-negative sums force equality.** If $\sum$(non-negative terms) $= 0$ then every term is 0. Choose the weights so the given hypotheses make the total vanish.

Where it usually goes wrong

- **Double counting an object that is not counted the same number of times in both directions.** Always ask: "how many times does a typical object contribute to each side?"
- **Bounding the wrong side.** A lower bound on T is useless if the other estimate is also a lower bound; make sure the two estimates point at each other.
- **Using convexity without checking the domain.** In IMO 1998 P2 the convexity step needs b odd to get the exact minimum $\frac{(b-1)^2}{4}$; with b even the answer changes.
- **Assuming points in general position are in convex position.** In the Iran 2010 problem, "no three collinear" gives at most 4 points on the two parallel lines; that count is the whole

geometric input.
- **Losing track of unordered vs. ordered.** Triples of the form (X, Y, Z) and (Z, Y, X) are different; $\{u, v\}$ and $\{v, u\}$ are the same. Fix a convention and divide accordingly — once.

Part III Deep Combinatorics

7. Extremal Combinatorics

Snapshot

Extremal combinatorics is "the combinatorics of superlatives": find or characterise the maximum, minimum, best or worst object satisfying some constraints. **Ch. 7** is the first of three chapters that take the reader into the heart of the subject, and it assumes everything before it: invariants, the extremal principle, induction and double counting all reappear as tools.

The chapter's own new methods are:

- **Injections and bijections** — to compare sizes of carefully chosen sets.
- **The alternating chains technique** — markings with no two consecutive, and its generalisation to chains in a graph.
- **Exploiting board structure** — sequential placement, layered colourings.
- **The classification method** — partition into k classes, each of which already satisfies the requirement, then pigeonhole.
- **Induction in set combinatorics** — Kleitman's lemma and the Sunflower lemma.

The essential idea

An extremal problem always has two halves, and the book is emphatic that both are needed:

(i) a construction achieving the claimed extreme value, and
(ii) a proof that nothing better is possible.

The construction is usually easy (and is often found first, by experiment). The proof of optimality is where the techniques live. And the most common optimality engine is: *build a map out of the hypothesised largest object and prove it is injective.*

Alternating chains in one line. Among n points in a row, a set with no two consecutive points has size at most $\lceil n/2 \rceil$; on a circle, at most $\lfloor n/2 \rfloor$. If the object has to be *edgeless* in some auxiliary graph, decompose that graph into chains or cycles and add up the per-chain maxima.

Tools and results

(a) Injections and bijections

The scheme: to prove $|U| \le |V|$, exhibit an injection $U \to V$. The steps are always the same — define the map using maximality or a structural choice, then prove injectivity using the hypothesis.

- **APMO 2008.** Groups of 3, any two groups sharing at most one student, 46 students. There is a set of at least 10 students containing no group. *Take the largest such set S.* For every $X \notin S$ there is a group $\{X, A, B\}$ with $A, B \in S$ (else add X). Map $X \mapsto \{A, B\} \in \binom{S}{2}$. This is injective: two X's mapping to the same pair would give two groups meeting in two students. Hence $46 - |S| \le \binom{|S|}{2}$, and $|S| = 9$ fails ($37 > 36$), so $|S| \ge 10$.
- **ISL 1988 (separating and covering).** t subsets of an n-set must separate every pair and cover every element. Encode each element by its t-bit membership string: separation means the n strings are distinct, covering means none is the all-zero string. So $n \le 2^t - 1$, i.e. $t = \lfloor \log_2 n \rfloor + 1$, and the bound is achieved by reversing the encoding. *The binary-string encoding is the seed of algebraic combinatorics, where these strings become vectors over $\mathbb{F}_2$.*

- **IMO 2006 P2.** A regular 2006-gon is dissected into triangles; a *good* triangle is isosceles with two odd sides (a side of the polygon counts as an odd diagonal). The maximum number of good triangles is $n = 1003$. *Map each side XY of the polygon to the good triangle ABC determined by the smallest good arc containing X and Y.* Each side maps to at most one triangle, and one shows each good triangle is hit by at least two sides: if not, the sides inside arc DE pair up by reflection across the perpendicular bisector of $D'F'$, but arc DE has an odd number of sides and cannot be perfectly paired. *Parity beats cleverness.*

(b) Alternating chains

- **ISL 1990 (red points on a circle).** $2n-1$ points on a circle, k coloured red; a colouring is *good* if some two red points have exactly n points of E in one of the arcs between them. Bad = no two red points joined by an edge A_iA_{i+n+1}. Those edges decompose the $2n-1$ points into $\gcd(n+1, 2n-1)$ cycles, and $2n-1 = 2(n+1)-3$ makes this gcd equal to 3 if $3 \mid n+1$ and 1 otherwise. A cycle of length ℓ takes at most $\lfloor \ell/2 \rfloor$ red points. Hence the answer is $k = n$ when $3 \nmid n+1$, and $k = 3\lfloor (2n-1)/6 \rfloor + 1$ otherwise.
- **USAMO 2008 P3.** The lattice region $S_n = \{(x, y) : |x| + |y + \frac{1}{2}| \le n\}$ cannot be partitioned into fewer than n paths. Colour each row alternately red and black starting and ending with red: neighbouring points then have opposite colours *except* across the middle two rows. Any path, split at each pair of consecutive red points, contains at most one more red than black point; there are exactly n such red pairs, so m paths split into at most $m+n$ pieces. As S_n has $2n$ more red than black points, $m+n \ge 2n$.
- **USAMO 2009 P2.** Largest subset of $\{-n, \dots, n\}$ containing no three elements (not necessarily distinct) summing to 0: n if n is even, $n+1$ if n is odd, achieved by the odd numbers. For the upper bound, let j be the smallest absolute value in S. If $-j \notin S$, pair up $n-2j+1$ pairs summing to $-j$, each contributing at most one element. If $-j \in S$, join $a, b \in T$ when $a + b = \pm j$: every element has degree 1 or 2 in this graph, so T decomposes into $2j$ disjoint chains, and a chain of length ℓ_i contributes at most $\lceil \ell_i/2 \rceil$. Summing gives $n+1$, refined to n for even n by exhibiting a chain of even length.

(c) Exploiting board structure

- **Sequential placement / the "time" trick (USAMO 1999 P1).** Checkers on an $n \times n$ board such that (a) every empty square is adjacent to a checker and (b) the checker set is connected. Then at least $\frac{n^2-2}{3}$ checkers. Because of (b) we may *imagine the checkers being placed one at a time, each adjacent to the previous one.* The first checker makes at most 5 squares good; each later checker makes at most 3 new squares good. So k checkers make at most $3k+2$ squares good, and all n^2 must be good. *Introducing an order into a static configuration is a powerful move*; the argument only uses "every square has at most 4 neighbours", so it generalises to graphs of maximum degree Δ.
- **Layered colourings (IMO 1999 P3).** For even $n = 2k$, the minimum number of marked cells so that every cell (marked or not) has a marked neighbour is $k(k+1)$. Colour the board in layers so that *every* square, black or white, has exactly two black neighbours; there are $2k(k+1)$ black squares, so at least $k(k+1)$ must be marked. Alternate black squares within each layer achieve it. *Colourings are how you exploit the structure of a board*; the right colouring turns a local condition into a global count.

(d) The classification method

To show that a large subset $S' \subseteq S$ with a property exists: partition $S = S_1 \cup \cdots \cup S_k$ so that *every* class has the property; pigeonhole gives a class of size $\ge |S|/k$.

ISL 2001 C6. A *balanced sequence* has n zeros and n ones, and two balanced sequences are *neighbours* if one symbol can be moved to another position. There is a set S of at most $\frac{1}{n+1}\binom{2n}{n}$ balanced sequences such that every balanced sequence is a neighbour of a member of S. Define

$f(A) = \sum$(positions of the 1s) mod $(n+1)$ and split the balanced sequences into $n+1$ classes by f. Each class is a dominating set: moving the first 1 to just after the k-th zero changes f by exactly k, and k can be chosen to hit any target residue. *Design the partition so that the required property becomes an algebraic congruence.*

(e) Two graph problems — the "worst case in the middle" heuristic

- **Swell colourings.** In a colouring of the edges of K_n where every triangle is either rainbow or monochromatic, at least $\sqrt{n}+1$ colours are needed. Take x_0, c_0 maximising the number N of c_0-coloured edges at x_0; then $Nk \geq n-1$. For any y joined to x_0 by a non-c_0 edge, the $N+1$ edges $yx_0, yx_1, \ldots, yx_N$ must be pairwise different and none of them c_0: hence $k \geq N+2$, and combining gives $k(k-2) \geq n-1$. *Either there is a large monochromatic structure, or there are many colours.*
- **Hats (Belarus 2001).** Friends/enemies, hats so that friends share a colour and enemies share none; the answer is $\lfloor n/2 \rfloor \lceil n/2 \rceil$. The *worst case is balanced bipartite* (few edges ⇒ no constraint, many edges ⇒ large cliques), which needs exactly $\lfloor n/2 \rfloor \lceil n/2 \rceil$ colours. Then strong induction: if the graph has fewer edges than that, colour each edge separately; if more, it contains a triangle (contrapositive of Mantel, **Ch. 6**); colour the triangle with one colour, spend at most $n-3$ colours on its neighbours, and induct on the remaining $n-3$ vertices. *When an extremal answer looks like a product of two nearly equal halves, the extremal example is bipartite.*

(f) Induction in the combinatorics of sets

Two lemmas first. For a family $\mathcal{F}$ of subsets of an n-set:

$$\textbf{Lemma 7.1:}\ A \cap B \neq \varnothing\ \forall A, B \in \mathcal{F} \implies |\mathcal{F}| \leq 2^{n-1},$$

$$\textbf{Lemma 7.2:}\ A \cup B \neq S\ \forall A, B \in \mathcal{F} \implies |\mathcal{F}| \leq 2^{n-1}.$$

Both are immediate: pair each subset with its complement; equality in 7.1 by taking all sets containing a fixed element, in 7.2 by taking all sets missing a fixed element.

Kleitman's lemma. If $\mathcal{F}_1, \mathcal{F}_2$ are downward closed and $\mathcal{F}_3$ upward closed, then

$$|\mathcal{F}_1 \cap \mathcal{F}_2| \geq \frac{|\mathcal{F}_1|\,|\mathcal{F}_2|}{2^n}, \qquad |\mathcal{F}_1 \cap \mathcal{F}_3| \leq \frac{|\mathcal{F}_1|\,|\mathcal{F}_3|}{2^n}.$$

Proof of the first: induct on n, splitting each family into the sets containing n and the sets not containing n, and observing that for downward-closed families the "containing n" part, with n deleted, is a subfamily of the "not containing n" part; Chebyshev's inequality then finishes. *Interpretation: imposing one constraint halves a family; imposing two independent constraints quarters it.*

Applications:

- **Iran TST 2008.** If $\mathcal{F}$ has 2^{n-1} subsets of an n-set and every three have non-empty intersection, then some element lies in *all* of them. If two sets meet in exactly one element, that element is in everything. Otherwise the family is 2-intersecting, and the induction shows $|\mathcal{F}| < 2^{n-1}$ — contradiction.
- **Corollary of Kleitman.** If every two sets of $\mathcal{F}$ intersect and no two cover S, then $|\mathcal{F}| \leq 2^{n-2}$: extend $\mathcal{F}$ downwards to D and upwards to U; $\mathcal{F} = U \cap D$, $|U| \leq 2^{n-1}$, $|D| \leq 2^{n-1}$, so $|\mathcal{F}| \leq 2^{n-2}$.
- **Sunflower lemma.** A *sunflower* with k petals and core X is a family $S_1, \ldots, S_k$ with $S_i \cap S_j = X$ for $i \neq j$. If every set has size s and $|\mathcal{F}| > s!(k-1)^s$, then $\mathcal{F}$ contains a sunflower with k petals. Induct on s: take a *maximal* family of pairwise disjoint sets; if there are $\geq k$ you are done, otherwise their union B has $\leq s(k-1)$ elements and meets every set of $\mathcal{F}$ (maximality); pigeonhole gives an element $x \in B$ in more than $(s-1)!(k-1)^{s-1}$ sets, delete x and apply induction. *Maximality ⇒ the union is a transversal; pigeonhole ⇒ a popular element.*

- **USA TST 2011 (stronger form).** Pairs (a, b) with $1 \le a < b \le 2^n$; if $|S| > n\,2^n$, then S contains $(a, c), (b, d), (a, d)$ for some $a < b < c < d$. Induct; if neither half of the range is rich enough, more than 2^n pairs cross the midpoint. There are at most 2^{n-1} "B-champions" and 2^{n-1} "A-champions", so some crossing pair (x, y) is neither — and that is exactly what produces the fourth point. *Champions are the book's name for a cleverly chosen extremal witness; the construction of the definition is reverse-engineered from the configuration you want.*

Landmark problems from Ch. 7

Problem	Method
Groups of 3, 46 students (APMO 2008)	Largest avoiding set + injection into pairs.
Separating/covering family (ISL 1988)	Binary membership strings; $n \le 2^t - 1$.
Good triangles in a 2006-gon (IMO 2006 P2)	Map sides to good triangles; odd arcs cannot be paired.
Red points on a circle (ISL 1990)	Edges A_iA_{i+n+1}; $\gcd(n+1, 2n-1)$ cycles; $\lfloor \ell/2 \rfloor$ per cycle.
S_n paths (USAMO 2008 P3)	Alternating row colouring; split paths at consecutive red pairs; $m + n \ge 2n$.
Zero-sum triples (USAMO 2009 P2)	Smallest absolute value j; pairs summing to $-j$, or chains with edges $a + b = \pm j$.
Checkers on a board (USAMO 1999 P1)	Sequential placement; $5 + 3(k-1)$ good squares.
Marked cells (IMO 1999 P3)	Layered colouring with every square having exactly two black neighbours.
Balanced sequences (ISL 2001 C6)	Partition by $f(A) \bmod n+1$; each class is dominating.
Swell colourings	Maximise $N(x, c)$; $Nk \ge n - 1$ and $k \ge N + 2$.
Hats (Belarus 2001)	Balanced bipartite is the worst case; triangle + induction.
Iran TST 2008 / Kleitman	Lemma 7.1 + induction; downward/upward closures and 2^{n-2}.
Sunflower lemma	Maximal disjoint subfamily, transversal union, pigeonhole, induction on s.
Pairs in $[1, 2^n]$ (USA TST 2011)	Induction + A-/B-champions.

Key takeaways

- **Always produce both halves:** construction and optimality proof. Experiment with small cases to guess the answer *and* to find the construction.
- **Maximality ⇒ injection.** This is the single most reused engine in the chapter.
- **Odd numbers cannot be paired.** Alternating-chain arguments often end with a parity contradiction.
- **Order a static configuration.** Place the objects one at a time; charge each one only for the new "good" squares it creates.
- **Colour to convert locality into counting.** Find a colouring in which each object sees a fixed number of a chosen colour.
- **Classification + pigeonhole** gives $|S|/k$ for free — so choose the partition to make the property automatic within a class.
- **Two constraints ⇒ a factor of 4.** Lemmas 7.1/7.2 halve; Kleitman's lemma combines them.

- **Sunflower:** maximal disjoint subfamily, then pigeonhole on its union.
- **Worst cases live in the middle.** Extremal graphs are often balanced or bipartite; extremal set families are often "half of everything".

Where it usually goes wrong

- **Proving the bound but not exhibiting the example** (or the reverse). Both are required for "find the maximum".
- **Injecting into the wrong set.** The codomain must be counted exactly; in APMO 2008 the codomain is *unordered* pairs of S, not ordered ones.
- **Forgetting the exceptions in a colouring.** In USAMO 2008 P3, neighbouring points have opposite colours *except across the middle two rows* — that exception is precisely what produces the n red pairs and drives the whole proof.
- **Using a chain decomposition without checking the chain lengths.** The bound $\lceil \ell/2 \rceil$ per chain depends on the chain being *maximal* and vertex-disjoint.
- **Claiming a partial classification is complete.** Kleitman's lemma needs *both* families downward closed (resp. one upward); the roles are not interchangeable.
- **Assuming maximal disjoint subfamilies are unique.** They are not — but the *sizes* are controlled, and that is all the proof uses.

8. Graph Theory

Snapshot

Ch. 8 is not a graph-theory course. It is a chapter about *modelling*: turning a problem about people, sets, matrices, tournaments or polygons into a statement about vertices and edges, and then applying a small number of powerful lemmas. It opens by recalling six results proved earlier in the book, develops dominating sets, spanning trees, the extremal principle in graphs, and Hall's marriage theorem, and closes with a section on graphs that appear where nothing suggests them.

The essential idea

The routine. When a problem mentions pairwise relations — friendship, acquaintance, divisibility, intersection, adjacency, visibility — build a graph or digraph whose vertices are the objects and whose edges are the relation, then ask:

is this a degree condition, a connectivity condition, a colouring condition, or a matching condition?

Most olympiad graph problems are one of those four in disguise. The second half of the chapter is the harder skill: choosing a graph whose *structure* (a cycle, a bipartition, a spanning tree) supplies the contradiction you need.

Tools and results

(a) The recalled toolbox

1. Maximum degree $\Delta \Rightarrow$ proper colouring with $\Delta + 1$ colours (**Ch. 1**).
2. Average degree $\bar{d} \Rightarrow$ an induced subgraph of minimum degree $\geq \bar{d}/2$; equivalently, E edges on V vertices give an induced subgraph of minimum degree $\geq E/V$ (**Ch. 1**).
3. Minimum degree $\geq n-1$ and T any n-vertex tree $\Rightarrow G$ contains a copy of T (**Ch. 2**).
4. Minimum degree $\geq \delta \Rightarrow$ a path with $\geq \delta + 1$ vertices (**Ch. 4**).
5. V splits into V_1, V_2 so that each vertex has at least as many neighbours across as inside (**Ch. 4**).
6. Every tournament has a Hamiltonian path (Redei, **Ch. 4**).

(b) Dominating sets

$D \subseteq V$ is a **dominating set** if every vertex is in D or adjacent to a vertex of D.

Lemma 8.1. *If G has no isolated vertices, it has a dominating set of size at most $|V|/2$.* Proof: use result (5) above; since every degree is ≥ 1, each of V_1, V_2 dominates the other side, so both are dominating sets, and one has $\leq |V|/2$ vertices.

Application (Russia 2001). $2n+1$ people, and for every group of n people some person among the remaining $n+1$ knows everyone in the group. Then somebody knows everybody. Suppose not; build the *complement* graph G (edges = "do not know each other"), which has minimum degree ≥ 1. By Lemma 8.1 G has a dominating set D of at most $(2n+1)/2$, i.e. at most n, vertices. Every vertex outside D has a non-acquaintance inside D, so the n people of D form a group with no universal acquaintance outside — contradiction. *Working in the complement is often what makes a "knows everybody" problem into a degree problem.*

(c) Spanning trees

Lemma 8.2. *Every finite connected graph has a spanning tree.* Proof: delete one edge from each cycle until no cycle remains; connectivity is preserved and the graph is finite, so the process stops at a spanning tree.

Two facts are all you need: spanning trees are *trees*, and they *exist*. They let you replace an arbitrary connected graph by a minimal connected subgraph, which is what makes rootings and cuts manageable.

- **ELMO SL 2011 C7.** Any t-vertex tree is a subgraph of every n-vertex graph with at least $(t-1)n$ edges. Apply (2) to get an induced subgraph with minimum degree $\geq E/V \geq t-1$, then (3). *Two recalled lemmas compose into a two-line solution to a hard-looking problem.*
- **ELMO SL 2011 C2.** A digraph with in-degree = out-degree = 2 at every vertex can be partitioned into 3 sets so that no vertex shares a set with both of its out-neighbours. Take the partition *maximising crossing edges*; if a vertex v shares a set with both out-neighbours, move v to the other set with fewer of its in-neighbours: this adds 2 crossing edges and destroys at most 1.

(d) The extremal principle in graph theory

The book marks the extremal step in bold in five examples; the pattern is the same each time — *choose the graph, cycle, component or counterexample that is extreme in some way, then show the extremality forbids the bad configuration.*

- **ISL 2004 C3.** Starting from K_n, repeatedly delete an edge of a 4-cycle; the least possible number of edges is n. The graph stays connected ($\geq n-1$ edges) *and* non-bipartite: if deleting AB from the 4-cycle $ABCD$ made it bipartite for the first time, then A, B would be on the same side while BC, CD, DA force C to be on A's side and on B's side simultaneously. So $\geq n$ edges; the construction with a triangle plus a star achieves n. *Guess → experiment → notice the invariant (an odd cycle always survives).*
- **Croatian TST 2011.** Among any 4 people there are either 3 mutual friends or 3 mutual strangers. Then the people split into a clique A and an independent set B. Take A to be a *largest clique.* If v_1v_2 were an edge inside B, pick $u_1 \in A$ non-adjacent to v_1 (exists by maximality of A); a short case check on the 4-set shows that either $A \cup \{v_1, v_2\} \setminus \{u_1\}$ or $A \cup \{v_2\}$ is a larger clique — contradiction.
- **Degree vectors (Havel–Hakimi).** If $v = (d_1 \geq \cdots \geq d_k)$ is graphical, so is the vector obtained by deleting d_1 and subtracting 1 from the next d_1 components (then re-sorting). *Choose the graph realising v that minimises the sum S of the indices of x_1's neighbours.* If x_1x_i were a non-edge and x_1x_j an edge with $i < j$, then since $d_i \geq d_j$ there is an x_t with x_ix_t an edge and x_jx_t a non-edge; swapping the edges x_1x_j, x_ix_t for x_1x_i, x_jx_t preserves all degrees and decreases S by $j-i$. Hence x_1 is adjacent to exactly $x_2, \ldots, x_{d_1+1}$, and deleting x_1 realises v_1. *The extremal condition is reverse-engineered from the conclusion you want.*

- **MOP 2008.** If the edges of K_n are coloured and no colour is used more than $n-2$ times, there is a rainbow triangle. Suppose not; let X be a largest monochromatic connected component and suppose (say) it is red. For $v \notin X$ and any red edge u_1u_2 inside X, both vu_1 and vu_2 are non-red, so they must have the *same* colour (else a rainbow triangle); hence v joins X by a single colour c, and $X \cup \{v\}$ is a larger connected component of colour c. So X is everything, forcing $\geq n-1$ edges of one colour — contradiction.
- **USAMO 2007 P4 (generalised).** A connected graph on V vertices with maximum degree D splits into two connected subgraphs each with at least V/D vertices. Delete edges while connectivity survives, reducing to a tree; root the tree at x chosen to *minimise the size of its largest subtree* T. Then $|T| \geq (V-1)/D$ (there are $\leq D$ subtrees) and $|T| \leq V/2$ (else re-root at the first vertex of T, contradicting minimality). Take T and $G \setminus T$. *The "induction on edges" is really the observation that the worst case is a tree.*

(e) Hall's marriage theorem

A family of sets has a **system of distinct representatives** (SDR) if one element can be chosen from each set, all distinct. The **marriage condition**: for every subfamily of k sets, the union has at least k elements. It is obviously necessary.

Theorem (Hall). *The marriage condition is sufficient.*

Proof sketch. Delete elements from the sets until no further deletion is possible without violating the marriage condition; call the result $A_1', \ldots, A_n'$. Claim: each A_i' is a singleton. Otherwise some A_1' contains distinct x, y; by minimality, deleting x violates the condition (giving a subfamily P and a set X with $|X| \leq |P|$) and deleting y violates it (giving Y, Q with $|Y| \leq |Q|$). Adding $|X| + |Y| = |X \cap Y| + |X \cup Y| \leq |P| + |Q|$ and applying the marriage condition to $X \cup Y$ and $X \cap Y$ yields the reverse inequality with a $+1$, a contradiction. *The technique: not just using a hypothesis, but using the* failure *of a hypothesis.*

Graph form. In a bipartite graph with vertex set $V_1 \cup V_2$, a **complete matching** of V_1 matches every vertex of V_1 to a distinct vertex of V_2. Hall's condition becomes: $|N(S)| \geq |S|$ for every $S \subseteq V_1$.

Canada 2006 P3. In an $m \times n$ array of non-negative reals with every row and column containing a positive entry, and with equal row-sum and column-sum whenever a row and column meet in a positive entry: prove $m = n$. Make a bipartite graph (rows vs. columns, edge = positive entry) and show a matching from rows to columns exists, so $n \geq m$; by symmetry $m \geq n$. If no matching existed, some set S of rows has all its positive entries in a set T of fewer than $|S|$ columns; comparing the total of the S-rows computed by rows and by columns gives a contradiction because all row sums are positive. *Arrays of numbers and bipartite graphs are the same object, and the translation is worth internalising.*

(f) Graphs where you would not expect them

- **Taiwan 2001.** n distinct subsets $A_1, \ldots, A_n$ of an n-set: some element x can be deleted from all of them while keeping them distinct. Build a graph on the A_i; for each element y, if $A_i \setminus \{y\} = A_j \setminus \{y\}$ pick one such pair and join it by an edge. If no good x existed, every element contributes at least one edge and no two elements contribute the same edge, so $|E| \geq n$; a graph with n vertices and $\geq n$ edges has a cycle $A_1A_2 \cdots A_kA_1$. Following the element x_1 around the cycle (each consecutive pair differs only in x_i) forces $x_1 \in A_1$ — contradiction. *The cycle is the contradiction engine.*
- **USA TST 2002.** $|S| = 2^n + 1$, f maps pairs to $\{0, \ldots, 2^n - 1\}$, and for every triple one of $f(x,y), f(y,z), f(z,x)$ equals the sum of the other two. Then some triple has all three values 0. Induction: it suffices to find $S' \subseteq S$ with $|S'| \geq 2^{n-1} + 1$ on which f is always even — then halve. Build a graph joining a, b when $f(a,b)$ is odd. The triple condition forces 0 or 2 edges among any three vertices, and a graph with this property is bipartite (take a shortest odd cycle and derive a shorter one). The larger side of the bipartition has $\geq 2^{n-1} + 1$ vertices and is independent — done. *"0 or 2 edges among any three vertices $\Rightarrow$ bipartite" is a lemma worth memorising; taking the* shortest *odd cycle is the standard way to prove it.*

- **ISL 2002 C6.** For even $n = 2m$ there is a permutation $(x_1, \ldots, x_n)$ of $(1, \ldots, n)$ with $x_{i+1} \in \{2x_i, 2x_i - 1, 2x_i - n, 2x_i - n - 1\}$ (cyclically). Build a directed graph on $1, \ldots, m$ where vertex i has outgoing edges $2i - 1, 2i$ and incoming edges $i, i + m$. Every vertex has in-degree = out-degree, and a short induction shows the digraph is weakly connected; hence it has an *Eulerian circuit*, and reading off the successive edges gives the permutation. *Eulerian circuits are the tool for "arrange these objects cyclically so that consecutive ones satisfy a relation".*

Landmark problems from Ch. 8

Problem	Method
Lemma 8.1 (dominating set)	Apply the max-cut partition; no isolated vertices makes both sides dominate.
$2n + 1$ people (Russia 2001)	Complement graph; dominating set of size $\leq n$.
Tree in a dense graph (ELMO SL 2011 C7)	Compose recalled results (2) and (3).
Digraph, in = out = 2 (ELMO SL 2011 C2)	Partition maximising crossing edges; move a bad vertex.
Delete edges of 4-cycles (ISL 2004 C3)	Connectivity plus non-bipartiteness; answer n.
Clique/independent split (Croatian TST 2011)	Largest clique; a 4-set case check.
Degree vectors (Havel–Hakimi)	Minimise the index sum S of x_1's neighbours; an edge swap decreases S.
Rainbow triangle (MOP 2008)	Largest monochromatic connected component; every outside vertex attaches by one colour.
Split into two connected parts (gen. USAMO 2007 P4)	Reduce to a tree; root to minimise the largest subtree.
Hall's theorem	Minimal deletion; use the *failure* of the condition.
$m \times n$ array (Canada 2006 P3)	Bipartite rows-columns graph; Hall; conclude $m = n$.
n subsets of an n-set (Taiwan 2001)	Graph with $\geq n$ edges has a cycle; chase an element around it.
f on pairs (USA TST 2002)	Parity graph, 0 or 2 edges per triple, bipartite; shortest odd cycle.
Permutation from doubling (ISL 2002 C6)	Eulerian circuit in a digraph with in = out degree.

Key takeaways

- **Model first, prove second.** The right graph usually makes the problem look like something you already know.
- **Complement when the hypothesis is negative.** "Nobody knows everybody" becomes "minimum degree ≥ 1".
- **Spanning trees reduce a graph to a minimal connected skeleton** — use them whenever you need to induct or root.
- **Maximise/minimise a count to force a structure.** Maximum crossing edges ⇒ balanced cuts; largest monochromatic component ⇒ everything joins it; largest clique ⇒ independent remainder.
- **Edge swaps with an extremal potential.** The Havel–Hakimi proof is the model: define S, swap to decrease S, contradict minimality.
- **Hall's theorem is the membership criterion for matroid-like structures.** Use it

whenever the question is "can we assign distinct representatives?".
- **Eulerian circuits solve cyclic-arrangement problems.** Equal in- and out-degrees plus weak connectivity is all you need.
- **Shortest odd cycle.** Any "prove bipartite" argument should try it.

Where it usually goes wrong

- **Forgetting the connectivity hypothesis** when applying Lemma 8.1 or Lemma 8.2.
- **Using Hall's condition on the wrong side.** It must hold for *every* subset S of one side; one-sided checking is a common slip, and in the array problem you must also use positivity of the row sums.
- **Confusing "matching" with "perfect matching".** A complete matching of V_1 does not require all of V_2 to be used.
- **Extremal choice that is not maintained.** When re-rooting a tree or moving a vertex between parts, verify the extremal property still holds — that contradiction is the point.
- **Building a graph whose edges are "objects" instead of "relations".** In Taiwan 2001 the vertices are the sets and the *elements* label the edges; the counting of edges then comes from the number of elements.
- **Assuming a graph is bipartite from a 0-or-2-edges condition without a proof.** The proof needs the shortest odd cycle argument.

9. The Probabilistic Method

Snapshot

The probabilistic method proves results in combinatorics by introducing randomness *into the proof*. To show that an object with a property exists, you design a random process whose output is such an object, and show that with positive probability the output is good — or even just that the *expected* number of failures is less than 1.

The chapter is organised by method, one section each:

- **Linearity of expectation** and the pigeonhole property of expectation.
- **Mutually exclusive events** — the union bound in the sharp form $\sum \mathbb{P}[E_i] \leq 1$ when the E_i are pairwise disjoint.
- **Bounding violated conditions** — if $\mathbb{E}[\#\text{failures}] < 1$, some object fails at nothing.
- **Concentration bounds** — Markov, Chebyshev, Chernoff.
- **The Lovász Local Lemma** — when the failures are many but only locally dependent.

The essential idea

The pigeonhole property of expectation. A random variable cannot always be above its mean, and cannot always be below it:

$$\text{if } \mathbb{E}[X] = \mu \text{ then } \exists\, \omega : X(\omega) \geq \mu \text{ and } \exists\, \omega : X(\omega) \leq \mu.$$

This is the whole trick. You never need to exhibit the good object; you only need to show that the *average* object is good enough.

Two standard minimisation devices. After computing an expectation with a free parameter p, choose p to minimise it (calculus, or the AM–GM / Jensen inequality), then conclude. And remember: **linearity of expectation needs no independence**; concentration bounds and the LLL *do*.

Tools and results

(a) Linearity of expectation

- **The first-moment method.** If the expected number of good objects is μ, then some outcome has at least μ good objects. *Application (Szele's theorem):* in a random tournament (each game decided by a coin flip), a fixed permutation of the n vertices forms a Hamiltonian path with probability $2^{-(n-1)}$; there are $n!$ permutations; so the expected number of Hamiltonian paths is $n!/2^{n-1}$, and some tournament has at least that many.
- **Weighted counting to build a large subgraph with no K_{k+1} (gen. MOP 2010).** Take a random permutation $v_1, \dots, v_n$; select v_i iff it is among the first k of $\{v_i\} \cup N(v_i)$. Then every selected vertex has at most $k-1$ selected neighbours, so the selected set induces no K_{k+1}. Vertex v_i is selected with probability $\frac{k}{d_i+1}$, so the expected size is $\sum_i \frac{k}{d_i+1}$, and Jensen (convexity of $\frac{1}{x+1}$) gives

$$\mathbb{E}|H| \;\geq\; \frac{nk}{\bar{d}+1}, \qquad \bar{d} = \frac{2E}{n}.$$

  Setting $k=1$: **every graph has an independent set of size at least** $\sum_i \frac{1}{d_i+1} \geq \frac{n^2}{2E+n}$ (this is the Caro–Wei / Turán bound). A tree ($E = n-1$) therefore has an independent set larger than $n/3$.
- **Random selection, then repair (dominating sets).** Choose each vertex with probability p; let T be the set of vertices with no chosen neighbour. Then $S \cup T$ dominates, $\mathbb{E}|S| = np$, and $\mathbb{E}|T| \leq n(1-p)^{d+1} \leq ne^{-p(d+1)}$. Minimising $np + ne^{-p(d+1)}$ at $p = \frac{\ln(d+1)}{d+1}$ gives

$$\gamma(G) \;\leq\; \frac{n\,(1+\ln(d+1))}{d+1}$$

  for a graph of minimum degree $d > 1$. *The pattern: pick randomly, then patch up whatever got missed.*
- **Choose the parameter to balance two kinds of failure (USAMO 2010 P6).** 68 pairs of non-zero integers on a board, never both (k,k) and $(-k,-k)$. Erase integers so that no two erased sum to 0; maximise the guaranteed number of pairs touched. For each $k > 0$ delete all copies of k with probability p, else all copies of $-k$ (with $p > \frac{1}{2}$). A pair (k,k) scores with probability p; $(k,-k)$ always scores; (a,b) with $b \neq \pm a$ scores with probability $1 - \mathbb{P}[a \text{ survives}]\mathbb{P}[b \text{ survives}] \geq 1 - p^2$ (each survival probability is at most p, because $p > 1-p$). Hence the expectation is at least $68\min\{p, 1-p^2\}$, maximised at $p = \frac{\sqrt{5}-1}{2}$ where it exceeds 42. So 43 points are guaranteed, and 43 is the answer. *The optimisation is the problem.*
- **Two inequalities finish the job.** ISL 2012 C7 (disjoint equal-value chords): the graph G_c of chords of value c has an independent set of size at least $\sum_v \frac{1}{d(v)+1}$; averaging over the $4n-3$ values of c and summing over chords (degrees are at most i, with $2n$ chords per degree level) gives an average exceeding $\ln(n+1)$, hence more than 100. The reason this works is exactly example 2's bound.
- **Random winners (biclique covering).** Flip a coin for the winner of each match; a student who wins all matches he plays is a *champion*. There are at most α champions (any $\alpha+1$ contain a friendly pair, which cannot both be champions). But student k is a champion with probability 2^{-m_k}, so by AM–GM the expected number of champions is $\sum_k 2^{-m_k} \geq n\,2^{-S/n}$. Hence $n2^{-S/n} \leq \alpha$, i.e. $S \geq n\log_2(n/\alpha)$.

(b) Mutually exclusive events

If $E_1, \dots, E_n$ are pairwise mutually exclusive then $\sum_i \mathbb{P}[E_i] = \mathbb{P}[\bigcup E_i] \leq 1$. *Combinatorially this says: construct a family of disjoint events; the sum of their probabilities is at most one.*

- **LYM inequality.** For an antichain $A_1, \dots, A_s$ in the power set of $[n]$: $\sum_i \binom{n}{|A_i|}^{-1} \leq 1$. Take a random permutation of $[n]$ and let E_i be the event that A_i occupies the first $|A_i|$ positions. Distinct events cannot both occur (one set would contain the other), and $\mathbb{P}[E_i] = \binom{n}{|A_i|}^{-1}$.

- **Sperner's theorem (corollary).** Since $\binom{n}{\lfloor n/2\rfloor}$ is the largest binomial coefficient, LYM gives $s \le \binom{n}{\lfloor n/2\rfloor}$. Equality by taking all sets of size $\lfloor n/2\rfloor$.
- **Bollobás' theorem.** If $|A_i| = s$, $|B_i| = t$ and $A_i \cap B_j = \varnothing$ iff $i = j$, then $n \le \binom{s+t}{s}$. Take a random permutation of the union and let E_i be "every element of A_i precedes every element of B_i". Two E_i's cannot both occur (the sets would be too far apart), and $\mathbb{P}[E_i] = 1/\binom{s+t}{s}$.
- **Tuza's variant.** Under the weaker hypothesis $A_i \cap B_i = \varnothing$ and $(A_i \cap B_j) \cup (A_j \cap B_i) \neq \varnothing$ for $i \neq j$, one gets the weighted inequality

$$\sum_i p^{|A_i|}(1-p)^{|B_i|} \;\le\; 1 \qquad (0 < p < 1),$$

  by selecting each element independently with probability p; the corollary with $|A_i| = a$, $|B_i| = b$ is $n \le \binom{a+b}{a}$. *The shape of the desired inequality tells you the random experiment.*
- **Ice cream (inspired by China TST 1997).** n flavours, k cone sizes, any two flavours share a kid who buys them in different sizes; then the total number of ice creams $T \ge n \log_k n$. Label each kid by a random integer in $\{1, \dots, k\}$; the event "every kid eating flavour F eats it in the size matching their label" has probability $k^{-|F|}$, and the events for different flavours are mutually exclusive precisely because of the hypothesis. AM–GM converts $\sum k^{-|F|} \le 1$ into $T \ge n \log_k n$. *"How do you get a logarithm out of ice cream? The probabilistic method."*
- **Perimeter of a partition (ISL 2009 C4).** On a $2^m \times 2^m$ board partitioned into rectangles with the diagonal unit squares separate, the total perimeter is minimised at $(m+1)2^{m+2}$. Randomly select each rectangle with probability $\frac{1}{2}$; the events E_i = "all rectangles meeting row i are selected and none meeting column i is" are mutually exclusive, since E_i and E_j disagree about the rectangle covering (i, j). AM–GM on $\sum 2^{-(|R_i|+|C_i|)} \le 1$ gives the bound. The construction is recursive, splitting into four boards of side 2^{m-1}.

(c) Bounding violated conditions

Construct the object randomly; *show that the expected number of violated conditions is less than* 1; then some object violates none.

- **Erdős' theorem on tournaments.** A tournament has property P_k if every set of k players has a player who beats them all. In a random tournament, $\mathbb{P}[A_S] = (1-2^{-k})^{n-k}$ for the event "no outsider beats all of S" (the outsiders' outcomes are independent), so the expected number of bad k-sets is $\binom{n}{k}(1-2^{-k})^{n-k}$. If this is < 1, a tournament with P_k exists.
- **Taiwan 1997.** k-sets pairwise meeting in at most $k-2$ elements: there is a subset M with $\lfloor n/2\rfloor + 1$ elements containing no member of the family. First bound $|\mathcal{F}_k| \le \binom{n}{k-1}/k$ by double counting (each $(k-1)$-set lies in at most one member, each member contains k of them); then take a random m-element subset and bound the expected number of members it contains by $\binom{m}{k} \cdot |\mathcal{F}_k|/\binom{n}{k} < 1$.
- **Avoiding monochromatic progressions (inspired by ISL 1987).** A set of k terms in arithmetic progression is *boring* if it uses at most $l < k$ colours. Colour $1, \dots, n$ randomly with j colours. The expected number of boring sets is $A \cdot p$, where

$$A = \#\{k\text{-term APs in } [n]\} < \frac{n^2}{2(k-1)}$$

  (an AP is determined by its first two terms, and there are $n - (k-1)d$ choices of starting point for each common difference d), and $p = \mathbb{P}[\text{a fixed AP is boring}]$ is the probability that k independent random colours take at most l distinct values — exponentially small in $k - l$. The book's hypothesis is a square-root bound on n of the form $n < \left[c\,(k/l)^{\,k-l}\right]^{1/2}$ chosen exactly so that the polynomial factor A is beaten by the exponential decay of p, giving $Ap < 1$. Hence some colouring of $[n]$ contains no boring progression.
- **Kleitman–Spencer.** For $m \ge k^2 2^k \ln n$ there are m subsets of $[n]$ whose intersections with every k-set T realise all 2^k patterns. Random subsets: for fixed (T, T_1), each A_i realises the pattern with probability 2^{-k}, so the failure probability is at most 2^{-km}; multiply by the $\binom{n}{k}2^k$ pairs and use $m \ge k^2 2^k \ln n$ to make the product < 1.

(d) Concentration bounds

- **Markov.** For $X \geq 0$ and $a > 0$, $\mathbb{P}[X \geq a] \leq \mathbb{E}[X]/a$. *Intuition: not everyone can earn more than five times the average.*
- **Chebyshev.** $\mathbb{P}[|X - \mathbb{E}X| \geq a] \leq \mathrm{Var}(X)/a^2$ — Markov applied to $(X - \mathbb{E}X)^2$. *The spread is governed by the variance.*
- **Chernoff (simple form).** For $X = \sum_{i=1}^{n} X_i$ with X_i independent 0–1 variables and $\mathbb{E}X = \mu$:

$$\mathbb{P}[X > (1+\delta)\mu] \leq e^{-\mu\delta^2/3}, \qquad \mathbb{P}[X < (1-\delta)\mu] \leq e^{-\mu\delta^2/2} \qquad (0 < \delta < 1).$$

  Use Chernoff for relative deviations and when the variance is awkward; it needs independence.
- **Hypergraph discrepancy.** For a family of m sets each of size at most s, there is a red/blue colouring whose discrepancy is at most $2\sqrt{s\ln(2m)}$. Colour randomly; for a set S of size $t \leq s$, Chernoff gives $\mathbb{P}[\mathrm{disc}(S) > d] \leq 2e^{-d^2/4t} \leq 2e^{-d^2/4s}$, and the union bound over the m sets finishes.

(e) The Lovász Local Lemma

Degree of dependence. Draw a graph on the events $E_1, \ldots, E_n$ with an edge between E_i and E_j exactly when they are *not* independent; let d be the maximum degree.

Theorem (LLL). *If $\mathbb{P}[E_i] \leq p$ for all i and $ep(d+1) \leq 1$, then $\mathbb{P}[\bigcap_i \overline{E_i}] > 0$: with positive probability none of the events occurs.*

When to use it. The first-moment method needs $\mathbb{E}[\#\text{bad}] < 1$, i.e. roughly $np < 1$. The LLL needs only $p(d+1) \leq 1/e$, i.e. $np \lesssim n/(ed)$ — a factor of n/d better. *The price is that "bad" events must be locally dependent*: different vertices' conditions can only interact through shared neighbours. The presence of *only local information* in the statement (degrees, neighbourhoods, never the total number of vertices) is the signal to reach for the LLL.

Worked example (gen. Russia 2006). At a party, everyone has between δ and Δ friends. Colour the people with W wine types so that each person has $k+1$ friends with pairwise different wines. Randomly colour; let E_v be the bad event at v. Then

$$p \;\leq\; \binom{d_v}{k}\Big(\frac{1}{W}\Big)^k \;\leq\; \Big(\frac{e\,d_v}{k}\Big)^k \frac{1}{W^k},$$

and E_u is dependent with E_v only when u is a neighbour of a neighbour of v, so $d \leq \Delta^2 - \Delta$. Substituting p and d into the LLL condition $ep(d+1) \leq 1$ and solving for W gives exactly the lower bound on W stated as the hypothesis of the problem. *Note the signature of the method: the statement mentions only degrees and neighbourhoods, never the number of people — so the answer must come from a local lemma.*

Landmark problems from Ch. 9

Problem	Method
Szele's theorem	$n!/2^{n-1}$ expected Hamiltonian paths in a random tournament.
Induced K_{k+1}-free subgraph (gen. MOP 2010)	Random permutation, keep v_i if it is among the first k in its closed neighbourhood; Jensen.
Dominating sets	Random p-set, repair the uncovered vertices; optimise p.
USAMO 2010 P6 (43 points)	Randomise the sign choice; balance p against $1 - p^2$; answer 43.
ISL 2012 C7 (equal chords)	Independent sets in the chord-intersection graphs, averaged over values; $\ln(n+1) > 100$.
Biclique covering	Random winners; at most α champions; AM–GM.
LYM / Sperner	Random permutation; sets occupying initial segments are mutually exclusive.

Problem	Method
Bollobás' theorem	Random permutation; A_i before B_i.
Tuza's variant	Select each element with probability p; the inequality's shape dictates the experiment.
Ice cream	Random labels $\{1, \dots, k\}$; mutually exclusive flavour events; AM–GM gives $\log_k$.
Perimeter (ISL 2009 C4)	Random half of the rectangles; disjoint row/column events; answer $(m+1)2^{m+2}$.
Erdős on tournaments	Expected number of bad k-sets $= \binom{n}{k}(1-2^{-k})^{n-k} < 1$.
Taiwan 1997	Double-count $\lvert\mathcal{F}_k\rvert$, then a random m-subset.
Kleitman–Spencer	Random subsets realise all 2^k patterns w.h.p.; union bound.
Hypergraph discrepancy	Chernoff per set, union bound over m sets.
Wine colouring	Lovász Local Lemma; p from the degree, d from two-step neighbourhoods.

Key takeaways

- **To prove existence, randomise.** Build the object at random; the average object must be at least average.
- **Two failure modes = choose p to balance them.** The dominating-set bound and the USAMO 2010 P6 problem are both optimisation in disguise.
- **Choose the random experiment from the shape of the target inequality.** $p^{|A|}(1-p)^{|B|}$ asks for independent selection with probability p.
- **First moment:** expected number of failures $< 1 \Rightarrow$ zero failures achievable. **LLL:** use it when failures are many but local.
- **AM–GM and Jensen are the algebra of the method.** They turn $\sum$ of a convex function into $n\times$ the value at the mean.
- **Independence:** not needed for linearity of expectation, needed for Chernoff and for the LLL's dependence graph.
- **Look for local, neighbourhood-only information** in a problem statement — it is the signature of the LLL.
- **Concentration upgrades existence into abundance.** If the random variable concentrates, then not only does a good object exist, but *most* objects are good.

Where it usually goes wrong

- **Using union bound when events can be disjoint in the wrong way.** For mutually exclusive events, $\sum \mathbb{P}[E_i] \le 1$ is sharp; for general events use $\sum \mathbb{P}[E_i]$ as an upper bound on the union, not on the complement.
- **Forgetting that "expected number of failures < 1" is a strict inequality.** $= 1$ gives nothing.
- **Treating dependent events as independent** in Chernoff or in a direct product computation. In Erdős' tournament theorem, the independence over outsiders is real and must be stated.
- **Applying the LLL without bounding the degree of dependence correctly.** Two local events at u, v are dependent when they share a neighbour; the count is $\Delta^2 - \Delta$, not Δ.
- **Confusing the two different meanings of "success".** In the biclique problem the random choice is the *winner*, not the partition — the problem says nothing about winners, so we are free to choose them.
- **Optimising p outside its allowed range.** Several problems restrict $p \in (1/2, 1)$; the optimum must lie in the interval.

- **Expectation $\neq$ existence of a large *simultaneous* outcome.** The pigeonhole property gives one outcome above the mean for one statistic; if you need several statistics simultaneously, extra work is required.

Part IV Synthesis and Practice

10. How the nine chapters hang together

The single most useful thing to take away from the book is that its nine chapters form *one* method with nine faces. The same problem can often be solved in two or three of them, and choosing the right face is the skill.

The dependency web

Link	What actually transfers
Ch. 1 → **Ch. 3**	A monovariant does double duty. In **Ch. 1** it tells *you* which move to make; in **Ch. 3** it tells you what the *given* process must do. The proofs are the same; only the quantifier over moves changes.
Ch. 1 → **Ch. 5**	The "good/bad position" device of **Ch. 1** becomes positional analysis in **Ch. 5**: a good position is one from which you can reduce the opponent's options. Symmetry and pairing strategies are just invariants of a game.
Ch. 2 → **Ch. 4**	Induction is constructive in **Ch. 2** and existential in **Ch. 4**. The top-down form (delete objects to reach the hypothesis) exists precisely because in an existence proof you are free to *choose* which objects to remove.
Ch. 3 → **Ch. 4**	Bound-the-steps arguments in **Ch. 3** are the same per-object counts that, in **Ch. 4**, produce injections: "object x can be involved at most $f(x)$ times" is an inequality on a sum.
Ch. 4 → **Ch. 5**	Existential reasoning about games. Russia 2011 (the $N > n^2$ stones) proves that Alice *has* a winning strategy without exhibiting it: suppose Bob wins, use pigeonhole on his replies, derive a contradiction. This is the **Ch. 4** extremal principle wearing a game's clothes.
Ch. 4 → **Ch. 7**	The optimal assumption is the backbone of **Ch. 7**: take the largest set avoiding the family, then extract an injection from maximality.
Ch. 6 → **Ch. 7**	Every extremal *bound* in **Ch. 7** is proved by **Ch. 6** double counting. Injections count two things; alternating chains count per-chain maxima; the classification method is a counting argument over classes.
Ch. 6 → **Ch. 8**	Incidence matrices *are* bipartite graphs. Row sums and column sums are degrees; a matching is a set of 1s with distinct rows and columns. Hall's theorem is therefore a statement about matrices as much as about graphs.
Ch. 7 → **Ch. 8**	Extremal graph theory is the intersection: Mantel's bound (no triangle $\Rightarrow E \leq n^2/4$) is used *inside* the hats problem of **Ch. 7**; Turán's theorem generalises it; both need **Ch. 6**.
Ch. 8 → **Ch. 9**	Graphs are the natural home of the probabilistic method: the max-cut partition, the dominating set bound, the induced K_{k+1}-free subgraph and the LLL all live on graphs, and the LLL's "degree of dependence" is literally a graph degree.

Link	What actually transfers
Ch. 9 → Ch. 4	The probabilistic method is the most powerful *existential* method of all: it proves existence when no construction is available, which is exactly the problem Ch. 4 set out to solve.
Ch. 3 → Ch. 9	"Expected number of failures < 1" is the probabilistic analogue of the monovariant argument: bound a counting quantity, conclude something about the process.

Three families that recur under different names

- **Extremal principle.** Ch. 1 (look at the extreme element), Ch. 3 (earliest bad configuration), Ch. 4 (largest set with the property), Ch. 5 (maximise/minimise in a game), Ch. 7 (largest avoiding set), Ch. 8 (longest path, largest clique, largest monochromatic component, minimal index sum). If you internalise one idea from this book, make it this one.
- **Injections from maximality.** "Adding x would break the property" $\Rightarrow$ x has a unique witness inside the maximal object $\Rightarrow$ an injection. This is the engine of APMO 2008, Italy TST 1999, the Sunflower lemma, and the pebble-chain argument of ISL 2001 C7.
- **Weighting.** Assign a weight so that a move has a controlled effect. 2^{-i} for halving processes, mod$n$ weights for cyclic processes, p and $1-p$ for probabilistic selection, $(-1)^t$ for parity-sensitive labels. Whenever a move is "double", "halve", "pass along a circle", or "flip", ask what weight linearises it.

11. Master reference: the lemmas worth memorising

Result	Statement and use
Greedy colouring (Ch. 1)	Max degree $\Delta \Rightarrow (\Delta+1)$-colouring; used whenever a colouring must exist.
Induced subgraph (Ch. 1)	E edges, V vertices $\Rightarrow$ an induced subgraph of minimum degree $\geq E/V$.
Dense graph contains trees (Ch. 2)	$\geq (t-1)n$ edges on n vertices $\Rightarrow$ every t-vertex tree appears.
Redei (Ch. 4)	Every tournament has a Hamiltonian path.
Max-cut partition (Ch. 4)	$V = V_1 \sqcup V_2$ with every vertex having at least as many cross-neighbours as internal ones.
Longest path (Ch. 4)	Minimum degree $\geq \delta \Rightarrow$ a path on $\geq \delta+1$ vertices.
Discrete continuity (Ch. 4)	$\lvert a_{i+1}-a_i\rvert \leq 1 \Rightarrow$ every integer between a_1 and a_n is attained.
Hostile neighbours (Ch. 4)	Two types in a cyclic sequence $\Rightarrow$ an adjacent opposite-type pair.
Incidence identity (Ch. 6)	$\sum_i \lvert A_i\rvert = \sum_j d_j$; the source of almost every set-system inequality.
Degree identities (Ch. 6)	$\sum d_i = 2E$; $\sum d_i^2 \geq 4E^2/n$; $\sum \binom{d_i}{2} \geq \frac{2E^2}{n} - E$; $\sum_i \sum_{j\sim i} d_j = \sum d_i^2$.
Tournament identity (Ch. 6)	$\sum w_i = \sum \ell_i = \binom{n}{2}$ and $\sum \binom{w_i}{2} = \sum \binom{\ell_i}{2}$.
Mantel (Ch. 6)	Triangle-free $\Rightarrow E \leq \lfloor n^2/4 \rfloor$, equality for balanced complete bipartite.
Triangle count (APMO 1989)	$\geq \frac{k(4k-n^2)}{3n}$ triangles when $\lvert E\rvert = k$.
No C_4	$E \leq \frac{n}{4}(1+\sqrt{4n-3})$.

Result	Statement and use
Sperner / LYM (**Ch. 9**)	Antichain: $\sum \binom{n}{\lvert A_i\rvert}^{-1} \le 1$, hence $s \le \binom{n}{\lfloor n/2\rfloor}$.
Bollobás (**Ch. 9**)	$A_i \cap B_j = \varnothing \iff i = j \Rightarrow n \le \binom{s+t}{s}$.
Tuza variant (**Ch. 9**)	$\sum p^{\lvert A_i\rvert}(1-p)^{\lvert B_i\rvert} \le 1$; corollary $n \le \binom{a+b}{a}$.
Lemmas 7.1/7.2 (**Ch. 7**)	Intersecting family $\le 2^{n-1}$; no two sets covering $S \le 2^{n-1}$.
Kleitman (**Ch. 7**)	Downward/downward: $\lvert\mathcal{F}_1 \cap \mathcal{F}_2\rvert \ge \lvert\mathcal{F}_1\rvert\lvert\mathcal{F}_2\rvert/2^n$; downward/upward: $\le$ the same. Corollary: intersecting and never covering $\Rightarrow \le 2^{n-2}$.
Sunflower lemma (**Ch. 7**)	Sets of size s; more than $s!(k-1)^s$ of them $\Rightarrow$ a sunflower with k petals.
Lemma 8.1 (**Ch. 8**)	No isolated vertices $\Rightarrow$ a dominating set of size $\le \lvert V\rvert/2$.
Lemma 8.2 (**Ch. 8**)	Every finite connected graph has a spanning tree.
Hall (**Ch. 8**)	$\forall S \subseteq V_1 :\ \lvert N(S)\rvert \ge \lvert S\rvert \iff$ a complete matching of V_1 exists. Consequences: König (max matching = min vertex cover, bipartite), Dilworth, k-regular bipartite $\Rightarrow$ k perfect matchings.
Havel–Hakimi (**Ch. 8**)	The degree-vector reduction lemma, proved by minimising the neighbour-index sum.
Independence bound (**Ch. 9**)	$\alpha(G) \ge \sum_i \frac{1}{d_i+1} \ge \frac{n^2}{2E+n}$; a tree has $\alpha > n/3$.
K_{k+1}-free subgraph (**Ch. 9**)	An induced K_{k+1}-free subgraph on $\ge \frac{nk}{d+1}$ vertices exists.
Dominating set bound (**Ch. 9**)	Minimum degree $d > 1 \Rightarrow$ a dominating set of size $\le \frac{n(1+\ln(d+1))}{d+1}$.
Markov / Chebyshev / Chernoff (**Ch. 9**)	$\mathbb{P}[X \ge a] \le \mathbb{E}X/a$; $\mathbb{P}[\lvert X - \mathbb{E}X\rvert \ge a] \le \mathrm{Var}(X)/a^2$; $\mathbb{P}[\lvert X-\mu\rvert \ge \delta\mu] \le 2e^{-\mu\delta^2/3}$.
Lovász Local Lemma (**Ch. 9**)	If $\mathbb{P}[E_i] \le p$ with degree of dependence d and $ep(d+1) \le 1$, then $\mathbb{P}[\bigcap \overline{E_i}] > 0$.

12. Problem → method: a decision guide

Read the problem statement and look for these signals.

Signal in the problem	First thing to try
"Show that one can … by a sequence of moves"	Greedy + monovariant (**Ch. 1**); if the target is astronomically large, build a recursive generator.
"Feasible / optimal assignment"	Greedy with an exchange argument; matroid if the feasible sets are closed under subsets.
"Show it is impossible"	Invariant: a weight, a parity, a value mod m (**Ch. 1**, **Ch. 3**, **Ch. 4**).
"An operation is repeated until no more moves are possible"	Monovariant with values in $\mathbb{N}$ (**Ch. 3**); then find the terminal state by extremality.
"The final configuration is unique"	Earliest bad configuration + the move before it (**Ch. 3**).
"At most $f(n)$ moves", with f polynomial	Bound each object's contribution and sum; look for a recursive per-object bound (**Ch. 3**).
"At most $f(n)$" with $f = 2n - c$ or $\lceil\log\rceil$	Induction with $f(n) \le f(n-1) + c$, resp. $f(n) = f(n/2) + c$ (**Ch. 2**).
"As few questions / weighings / comparisons as possible"	Decision-tree lower bound by counting outcomes; upper bound by an encoding (binary labels) (**Ch. 2**).

Signal in the problem	First thing to try
"Prove that there exists ..." with no construction in sight	Extremal principle: largest/smallest object; derive an injection from maximality (**Ch. 4**, **Ch. 7**).
"Prove that there exists A such that for all B ..."	Optimal assumption: take the largest A, then prove it is big enough by contradiction (**Ch. 4**).
"Two types, arranged in a line or circle"	Hostile neighbours / discrete continuity (**Ch. 4**).
"Choose a starting point so that all prefix sums are positive"	Minimum prefix sum / cycle lemma (**Ch. 4**).
Two players, symmetric board	Symmetry, mirroring, or a pairing (**Ch. 5**).
Two players, "last to move wins"	P/N-positional analysis, working backwards from the terminal position (**Ch. 5**).
Two players with opposite goals (maker–breaker)	Maintain a structural invariant (e.g. $B_i + B_{i+2} \leq 1$) (**Ch. 5**).
An inequality about sets, graphs or points	Count a carefully chosen quantity in two ways (**Ch. 6**).
"At most" / "at least" in the hypothesis	The quantifier tells you which direction to bound in the double count (**Ch. 6**).
"No two of them are too similar"	Counting triples / inclusion–exclusion on degrees or intersections (**Ch. 6**).
"Find the maximum/minimum number of ..."	Construction plus optimality proof; the optimality is an injection or a chain argument (**Ch. 7**).
Objects with a fixed size, families of subsets	Lemmas 7.1/7.2, Kleitman, Sunflower (**Ch. 7**).
"Partition into classes" or "there exists a large subset"	Classification method + pigeonhole (**Ch. 7**).
Relations, friendships, divisibility, adjacency	Build a graph; ask whether it is a degree, connectivity, colouring, or matching condition (**Ch. 8**).
"Assign distinct representatives"	Hall's theorem (**Ch. 8**).
"Arrange cyclically so consecutive objects satisfy a relation"	Eulerian circuit (in-degree = out-degree) (**Ch. 8**).
"Arrange objects in a sequence with constraints"	Hamiltonian path / tournament reasoning (**Ch. 8**).
"Only local information" (degrees, neighbourhoods)	Lovász Local Lemma (**Ch. 9**).
"Show there exists a colouring / partition avoiding all bad configurations"	Random colouring + expected number of violations < 1 (**Ch. 9**); upgrade to LLL if the expected number is large.
"Roughly balanced" or "within a constant factor"	Chernoff / Chebyshev after a linearity-of-expectation start (**Ch. 9**).

13. Practice set: eight graded exercises

These are **not** from the book (the book has its own large exercise sets at the end of each chapter). They are original or classical exercises chosen to exercise the *methods* above across chapter boundaries. Provenance is labelled. Hints and full solutions are in Section 14 — try each problem for at least fifteen minutes before reading them.

E1. (Positional analysis; original.) A pile contains $n \geq 1$ stones. Two players alternately remove

1, 2 or 5 stones; the player who takes the last stone wins. Determine exactly which n the first player wins, and describe the winning strategy.

E2. (Invariants; classical.) The numbers $1, 2, \dots, n$ are written on a blackboard. A move replaces two of them, a and b, by the single number $|a - b|$. After $n - 1$ moves one number F remains. Show that F is determined modulo 2 by n alone, and determine for which n the number F is odd.

E3. (Monovariant and the extremal principle; classical — also exercise 4 of Ch. 1.) Let P be a set of $2n$ points in the plane, no three collinear. Show that P can be partitioned into n pairs so that the n joining segments do not intersect.

E4. (Counting in two ways; original.) Let $\mathcal{F}$ be a family of 3-element subsets of an n-element set with the property that any two members of $\mathcal{F}$ have at most one element in common. Prove $|\mathcal{F}| \le \frac{n(n-1)}{6}$, and show the bound is attained when a Steiner triple system exists.

E5. (Hall's theorem; classical.) Let G be a k-regular bipartite graph with $k \ge 1$. Prove that the edges of G can be partitioned into k perfect matchings. Deduce that every doubly stochastic matrix with rational entries is a convex combination of permutation matrices.

E6. (Probabilistic method; classical.) Let G be a graph with n vertices and m edges. Prove that $V(G)$ can be split into two parts so that at least $m/2$ edges join the two parts. Then exhibit a graph for which no such split cuts more than $\frac{2}{3}m$ edges, and deduce what this says about the best possible universal constant.

E7. (Probabilistic method, first moment; adapted from Ch. 9, exercise 2.) Let $\mathcal{F}$ be a family of subsets of a finite set, where every member has at least $k \ge 1$ elements. Prove that if $|\mathcal{F}| < 2^{k-1}$ then the ground set can be coloured red/blue so that no member of $\mathcal{F}$ is monochromatic. Then consider the family of *all* k-element subsets of a $(2k-1)$-element set and explain what that example shows about how close 2^{k-1} is to the truth.

E8. (Information and efficiency; classical.) There are n coins, of which exactly one is counterfeit and is strictly heavier than a genuine coin; a balance scale reports which of two pans is heavier. Show that $\lceil \log_3 n \rceil$ weighings always suffice, and that no strategy can guarantee success in fewer.

14. Hints and full solutions

E1 — Subtraction game $\{1, 2, 5\}$

Hint. Compute the P-positions for $n = 0, 1, 2, \dots, 12$ by hand and look for a pattern. Note that $5 \equiv 2 \pmod 3$.

Solution. Call $n = 0$ a P-position (the player to move has lost). Then:

n	0	1	2	3	4	5	6	7	8	9	10	11	12
	P	N	N	P	N	N	P	N	N	P	N	N	P

The P-positions are exactly the multiples of 3. Proof: (i) from a multiple of 3, subtracting 1, 2 or 5 gives a number congruent to 2, 1 or 1 modulo 3 — never a multiple of 3, so every move from a P-position leads to an N-position; (ii) from $n \equiv 1$ remove 1 and from $n \equiv 2$ remove 2, reaching a multiple of 3. Hence **the first player wins iff** $3 \nmid n$, by removing $n \bmod 3$ stones at each turn and thereafter always restoring a multiple of 3. The presence of $5 \equiv 2 \pmod 3$ is what makes the modulus 3 rather than 6: the allowed moves cover both non-zero residues mod 3.

E2 — Parity invariant under $a, b \mapsto |a - b|$

Hint. Compare $a + b$ with $|a - b|$ modulo 2.

Solution. Since $|a - b| \equiv a - b \equiv a + b \pmod 2$ (as $-b \equiv b$), each move preserves the total of all numbers modulo 2. The sum of the initial entries is $1 + 2 + \cdots + n = \frac{n(n+1)}{2}$, so

$$F \equiv \frac{n(n+1)}{2} \pmod 2.$$

Hence F is odd iff $\frac{n(n+1)}{2}$ is odd, i.e. iff $n \equiv 1$ or $2 \pmod 4$. *Remarks.* (i) The invariant is only the parity — F itself is not determined; for $n = 4$ the final number can be 0 or 2. (ii) This is a purely *impossibility-style* invariant: it classifies which parities are achievable, and a separate construction shows the other values in the same parity class are reachable.

E3 — Non-crossing pairing of $2n$ points

Hint. Among all n-segment pairings, take one of *minimum total length*, and uncross it.

Solution. Among the finitely many ways to pair up the $2n$ points, choose a pairing $\mathcal{P}$ minimising the sum L of the n segment lengths. Suppose two segments AB and CD of $\mathcal{P}$ cross at an interior point X. Then $|AB| + |CD| = |AX| + |XB| + |CX| + |XD|$, while the triangle inequality gives

$$|AC| < |AX| + |XC|, \qquad |BD| < |BX| + |XD|,$$

with *strict* inequality because no three points are collinear (so A, X, C are not collinear). Adding,

$$|AC| + |BD| < |AB| + |CD|.$$

Replacing the crossing pair $\{AB, CD\}$ by $\{AC, BD\}$ therefore produces a pairing of strictly smaller total length — a contradiction. Hence $\mathcal{P}$ has no crossing segments. *The strictness needs general position; with collinearities the total length can tie and the argument needs a tie-break.*

E4 — Triples pairwise sharing at most one point

Hint. Count pairs (2-element subset, member of $\mathcal{F}$) containing it.

Solution. Let N be the number of pairs (P, T) where P is a 2-element subset of the n-element ground set and $T \in \mathcal{F}$ with $P \subseteq T$. Every member of $\mathcal{F}$ has exactly 3 two-element subsets, so $N = 3|\mathcal{F}|$. Conversely, if two distinct members of $\mathcal{F}$ both contained the same pair P, they would share two elements, contradicting the hypothesis. So every 2-element subset lies in at most one member, giving $N \le \binom{n}{2}$. Hence $3|\mathcal{F}| \le \binom{n}{2} = \frac{n(n-1)}{2}$, i.e. $|\mathcal{F}| \le \frac{n(n-1)}{6}$. Equality requires every pair to lie in exactly one triple — that is precisely a Steiner triple system on n points, which exists iff $n \equiv 1$ or $3 \pmod 6$.

E5 — k-regular bipartite graphs and doubly stochastic matrices

Hint. Hall's condition follows from counting edges leaving S.

Solution. Let the parts be A and B. Counting edges by their A-endpoint and by their B-endpoint gives $k|A| = k|B|$, so $|A| = |B|$. For any $S \subseteq A$, the number of edges with an endpoint in S is $k|S|$; all of them end in $N(S)$, and each vertex of $N(S)$ is incident with at most k such edges. Hence $k|S| \le k|N(S)|$, i.e. $|N(S)| \ge |S|$. By Hall's theorem G has a matching covering A, which — since $|A| = |B|$ — is perfect. Delete its $|A|$ edges: every vertex loses exactly one incident edge, so the remaining graph is $(k-1)$-regular bipartite. Induct on k. (The case $k = 0$ is vacuous, $k = 1$ is the Hall step itself.)

For the matrix statement: a doubly stochastic matrix with rational entries can be scaled to an integer matrix whose every row and column sum equals the same k; this is the biadjacency matrix of a k-regular bipartite multigraph, which splits into k perfect matchings, i.e. k permutation matrices $P_1, \dots, P_k$ with $A = \frac{1}{k}(P_1 + \cdots + P_k)$. *This is the Birkhoff–von Neumann theorem in the rational case, and it is Hall's theorem in disguise.*

E6 — A cut with at least half the edges

Hint. Colour the vertices independently at random and count crossing edges.

Solution. Colour each vertex red or blue independently with probability $\frac{1}{2}$. For each edge e, let $X_e = 1$ if its endpoints receive different colours and $X_e = 0$ otherwise; then $\mathbb{P}[X_e = 1] = \frac{1}{2}$ because the two colours are independent and equally likely. With $X = \sum_e X_e$, linearity of expectation gives

$$\mathbb{E}[X] = \sum_{e \in E} \mathbb{E}[X_e] = \frac{m}{2}.$$

By the pigeonhole property of expectation, some colouring satisfies $X \geq \mathbb{E}[X] = m/2$: that colouring is the required partition.

The constant $\frac{1}{2}$ is not the whole story. Take $G = K_3$, $m = 3$. Every bipartition puts 1 vertex on one side and 2 on the other, so it cuts exactly 2 edges: the largest cut is $2 = \frac{2}{3}m$. Hence any universal claim of the form "every graph has a cut of at least cm edges" must have $c \leq \frac{2}{3}$, and the probabilistic proof's $\frac{1}{2}$ is strictly weaker than what K_3 permits. Large disjoint unions of triangles show the ratio $\frac{2}{3}$ is approached but never exceeded in this family. *The lesson of the exercise is the shape of the argument — randomise, compute the expectation, invoke the pigeonhole property — not the numerical constant; sharpening the constant is a separate (and classical) question about maximum cuts.*

E7 — Two-colouring a sparse hypergraph

Hint. Colour each element independently; bound the probability that a fixed set is monochromatic, then use the union bound.

Solution. Colour every element red or blue independently with probability $\frac{1}{2}$. A fixed member S of size $s \geq k$ is monochromatic with probability

$$2 \cdot \left(\tfrac{1}{2}\right)^s = 2^{1-s} \;\leq\; 2^{1-k}.$$

If X denotes the number of monochromatic members, linearity of expectation gives

$$\mathbb{E}[X] \leq |\mathcal{F}|\, 2^{1-k} < 2^{k-1} \cdot 2^{1-k} = 1.$$

Since $X \geq 0$ is integer-valued, $\mathbb{E}[X] < 1$ forces $\mathbb{P}[X = 0] > 0$: some colouring has no monochromatic member.

How sharp is 2^{k-1}? Consider the family $\mathcal{F}_0$ of *all* k-element subsets of a $(2k-1)$-element set X. Every member has size exactly k, there are $\binom{2k-1}{k}$ of them, and $\mathcal{F}_0$ is *not* two-colourable: in any red/blue colouring of X the two colour classes have sizes summing to $2k-1$, so one class contains at least k elements, and any k of its elements form a monochromatic member of $\mathcal{F}_0$.

Now $\binom{2k-1}{k} > 2^{k-1}$ for every $k \geq 2$ (for $k = 2$, $3 > 2$; for $k = 3$, $10 > 4$; asymptotically $\binom{2k-1}{k}$ grows like $4^k/\sqrt{k}$ while 2^{k-1} merely doubles). So the example shows that 2^{k-1} is *not* the exact threshold: non-two-colourable families of k-sets exist with far more than 2^{k-1} members.

The honest conclusion is therefore one-sided: $|\mathcal{F}| < 2^{k-1}$ is a clean *sufficient* condition, obtained by the union bound, and it is not optimal. Replacing the crude count of edges by a control on the *maximum degree* of the hypergraph is exactly what the Lovász Local Lemma does (**Ch. 9**, exercise 5); the general question "how many k-sets force non-two-colourability?" is a classical problem of extremal set theory.

E8 — Counterfeit coin: $\lceil \log_3 n \rceil$ weighings

Hint. ($\leq$) Split into three groups as equal as possible and weigh two of them. ($\geq$) Count the possible outcomes.

Solution. *Upper bound.* Let $f(n)$ be the least number of weighings that suffice for n coins containing exactly one known-heavier counterfeit. Clearly $f(0) = f(1) = 0$ and $f(2) = 1$. For $n \geq 3$ put $a = \lfloor n/3 \rfloor \geq 1$ and $r = n - 2a \in \{0, 1, 2\}$, so that the coins split into groups of sizes a, a and r. Weigh the two groups of size a against each other:

- if one pan is heavier, the counterfeit lies in that group of a coins;
- if the pans balance, the counterfeit lies among the $r \leq 2$ remaining coins.

Hence $f(n) \leq 1 + \max(f(a), f(r))$. By induction $f(a) \leq \lceil \log_3 a \rceil$, and since $n \geq 3a$ we have $\lceil \log_3 a \rceil + 1 = \lceil \log_3(3a) \rceil \leq \lceil \log_3 n \rceil$. Also $f(r) \leq 1 \leq \lceil \log_3 n \rceil - 1$ because $n \geq 3$. Therefore $f(n) \leq \lceil \log_3 n \rceil$ for all n.

Lower bound. Any deterministic strategy is a ternary decision tree: each weighing has three possible outcomes (left heavier, right heavier, balance). A tree of depth w has at most 3^w leaves, and the n coins must lead to n distinguishable leaves (the identity of the counterfeit is the only unknown). Hence $3^w \geq n$, i.e. $w \geq \lceil \log_3 n \rceil$. *Both bounds are tight: this is a rare problem where the information-theoretic bound is achievable, which happens because each weighing can genuinely be arranged to have all three outcomes possible with the remaining candidates evenly split.*

15. Consolidated list of common mistakes

1. **Proving only half of an "iff".** Most "find all n" problems need one invariant (necessity) *and* one construction (sufficiency).
2. **A monovariant that is not monotone.** Check every legal move, including the ones you did not invent.
3. **Descent on a non-well-ordered quantity.** Descending positive reals need not terminate; you need integers or a fixed decrement.
4. **Maximality used without naming what breaks.** The sentence "then $S \cup \{x\}$ would also satisfy P" must actually appear.
5. **Injections that are not injective.** Usually the failure is that two different witnesses were assumed to be the same object.
6. **Wrong induction parameter** or an unverified base case.
7. **Wrong recursion for the target bound.** Linear bounds come from $f(n-1)$; logarithmic from $f(n/2)$.
8. **Multiplicity errors in double counting.** Each triangle is counted three times, each unordered pair twice.
9. **Losing the strictness in an inequality.** $\mathbb{E}[\#\text{failures}] < 1$ is needed, not ≤ 1.
10. **Independence assumed where it is not available**, or needed and not established — linearity of expectation needs none, Chernoff and the LLL do.
11. **Using the LLL without a correct dependence count.** Local events at u and v depend on each other when they share a *neighbour*; the count is $\Delta^2 - \Delta$, not Δ.
12. **Positional analysis on a game with cycles**, or a maker–breaker game where P/N-positions are meaningless.
13. **Hall's condition checked on only one side**, or only for the whole set.
14. **Colouring arguments with unverified exceptions.** The "middle two rows" exception in USAMO 2008 P3 is not a technicality — it is the proof.
15. **Guessing the extremal example before experimenting.** The book's own ISL 2004 C3 remark: the tempting guess $n-1$ is wrong; a small experiment reveals the surviving odd cycle.
16. **Forgetting that a graph-theoretic hypothesis needs connectivity** before applying Lemmas 8.1 and 8.2.

16. One-page recap

The whole book in twelve lines

1. **Build it.** Greedy plus an exchange argument; or greedy plus a monovariant that certifies termination.
2. **Weight it.** Manufacture a weight (powers 2^{-i}, residues modulo n, signs $(-1)^t$, probabilities p) so a move has a controlled effect.
3. **Recurse.** $f(n) \to f(n-1)$ for linear targets, $f(n) \to f(n/2)$ for logarithmic ones; count information when the question is "how few".
4. **Analyse, don't construct.** For a given process: invariant for impossibility, monovariant for termination, extremality for the terminal state.
5. **Alter the rules.** If a change is a relabelling, make it — bouncing ants become passing ants.
6. **Extremise.** Largest/smallest/longest, then extract an injection from maximality.
7. **Descend.** If P regenerates a smaller instance of itself in a finite set, P is empty.
8. **Cross a boundary.** Two types in a line give an adjacent opposite pair; steps of size ≤ 1 cross zero.
9. **Pair and mirror.** Symmetric replies, paired squares, and the fact that an even number of objects can always be paired.

10. **Classify positions.** P and N; always move to a P-position; a partial classification often suffices.
11. **Count twice.** One quantity, two countings, two hypotheses; the quantifiers tell you which way to bound.
12. **Randomise.** Linearity of expectation + pigeonhole; mutually exclusive events for exact inequalities; violated-conditions bound < 1; Chernoff and the LLL when you need concentration or locality.

If you remember only five lemmas

$$\sum_i d_i = 2E, \qquad \sum_i \binom{d_i}{2} \geq \frac{2E^2}{n} - E, \qquad \alpha(G) \geq \frac{n^2}{2E+n},$$

Hall: $\forall S \subseteq V_1,\ |N(S)| \geq |S| \iff$ a complete matching of V_1 exists,

LLL: $\mathbb{P}[E_i] \leq p,$ degree of dependence $d,$ $ep(d+1) \leq 1 \implies \exists$ outcome with no E_i.

Chapter summaries and attributions follow Pranav A. Sriram, *Olympiad Combinatorics* (August 2014).
The practice set in Section 13 is independent of the book.